31

Faraday's Law and Inductance

CHAPTER OUTLINE

31.1 Faraday's Law of Induction

31.2 Motional emf

31.3 Lenz's Law

31.4 Induced emf and Electric Fields

31.5 Generators and Motors

31.6 Eddy Currents

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ31.1 The ranking is E > A > B = D = 0 > C. The emf is given by the negative of the time derivative of the magnetic flux. We pick out the steepest downward slope at instant E as marking the moment of largest emf. Next comes A. At B and at D the graph line is horizontal so the emf is zero. At C the emf has its greatest negative value.

OQ31.2 **(i)** Answer (c). **(ii)** Answers (a) and (b). The magnetic flux is $\Phi_B = BA\cos\theta$. Therefore the flux is a maximum when $\vec{\mathbf{B}}$ is perpendicular to the loop of wire and zero when there is no component of magnetic field perpendicular to the loop. The flux is zero when the loop is turned so that the field lies in the plane of its area.

OQ31.3 Answer (b). With the current in the long wire flowing in the direction shown in Figure OQ31.3, the magnetic flux through the rectangular loop is directed into the page. If this current is decreasing in time, the *change* in the flux is directed opposite to the flux itself (or out of the page). The induced current will then flow clockwise around the loop, producing a flux directed into the page through the loop and

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opposing the change in flux due to the decreasing current in the long wire.

OQ31.4 Answer (a). Treating the original flux as positive (i.e., choosing the normal to have the same direction as the original field), the flux changes from

$$\Phi_{Bi} = B_i A\cos\theta_i = B_i A\cos 0° = B_i A$$

to $$\Phi_{Bf} = B_f A\cos\theta_f = B_f A\cos 180° = -B_f A.$$

$$\mathcal{E} = -\frac{\Delta\Phi_B}{\Delta t} = -\left[\frac{\left(-B_f A\right)-\left(B_i A\right)}{\Delta t}\right] = \frac{2\left(B_f + B_i\right)A}{\Delta t}$$

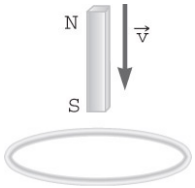
$$= 2\left[\frac{(0.060\text{ T})+(0.040\text{ T})}{0.50\text{ s}}\right]\left[\pi(0.040\text{ m})^2\right] = 2.0\times10^{-3}\text{ V}$$

$$= 2.0\text{ mV}$$

OQ31.5 Answers (c) and (d). The magnetic flux through the coil is constant in time, so the induced emf is zero, but positive test charges in the leading and trailing sides of the square experience a $\vec{\mathbf{F}} = q\left(\vec{\mathbf{v}}\times\vec{\mathbf{B}}\right)$ force that is in direction (velocity to the right) × (field perpendicularly into the page away from you) = (force toward the top of the square). The charges migrate upward to give positive charge to the top of the square until there is a downward electric field large enough to prevent more charge separation.

OQ31.6 Answers (b) and (d). By the magnetic force law $\vec{\mathbf{F}} = q\left(\vec{\mathbf{v}}\times\vec{\mathbf{B}}\right)$: the positive charges in the moving bar will feel a magnetic force in direction (velocity to the right) × (field perpendicularly out of the page) = (force downward toward the bottom end of the bar). These charges will move downward and therefore clockwise in the circuit. The current induced in the bar experiences a force in the magnetic field that tends to slow the bar: (current downward) × (field perpendicularly out of the page) = (force to the left); therefore, an external force is required to keep the bar moving at constant speed to the right.

OQ31.7 Answer (a). As the bar magnet approaches the loop from above, with its south end downward as shown in the figure, the magnetic flux through the area enclosed by the loop is directed upward and increasing in magnitude. To oppose this increasing upward flux, the induced current in the loop will flow clockwise, as seen from above, producing a flux directed downward through the area enclosed



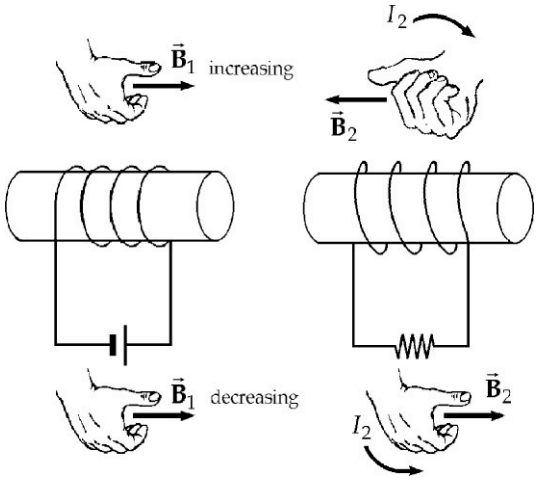
ANS. FIG. OQ31.7

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by the loop. After the bar magnet has passed through the plane of the loop, and is departing with its north end upward, a decreasing flux is directed upward through the loop. To oppose this decreasing upward flux, the induced current in the loop flows counterclockwise as seen from above, producing flux directed upward through the area enclosed by the loop. From this analysis, we see that (a) is the only true statement among the listed choices.

OQ31.8 Answer (b). The maximum induced emf in a generator is proportional to the rate of rotation. The rate of change of flux of the external magnetic field through the turns of the coil is doubled, so the maximum induced emf is doubled.

OQ31.9 **(i)** Answer (b). The battery makes counterclockwise current I_1 in the primary coil, so its magnetic field $\vec{\mathbf{B}}_1$ is to the right and increasing just after the switch is closed. The secondary coil will oppose the change with a leftward field $\vec{\mathbf{B}}_2$, which comes from an induced clockwise current I_2 that goes to the right in the resistor. The upper pair of hands in ANS. FIG. OQ31.9 represent this effect.



ANS. FIG. OQ31.9

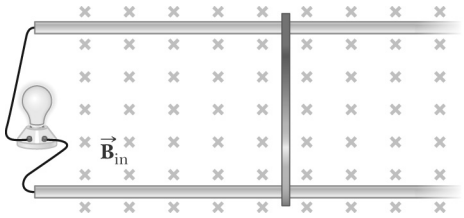
(ii) Answer (c). At steady state the primary magnetic field is unchanging, so no emf is induced in the secondary.

(iii) Answer (a). The primary's field is to the right and decreasing as the switch is opened. The secondary coil opposes this decrease by making its own field to the right, carrying counterclockwise current to the left in the resistor. The lower pair of hands shown in ANS. FIG. OQ31.9 represent this chain of events.

OQ31.10 Answers (a), (b), (c), and (d). With the magnetic field perpendicular to the plane of the page in the figure, the flux through the closed loop to the left of the bar is given by $\Phi_B = BA$, where B is the magnitude

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of the field and A is the area enclosed by the loop. Any action which produces a change in this product, BA, will induce a current in the loop and cause the bulb to light. Such actions include increasing or decreasing the magnitude of the field B, and moving the bar to the right or left and changing the enclosed area A. Thus, the bulb will light during all of the actions in choices (a), (b), (c), and (d).



ANS. FIG. OQ31.10

OQ31.11 Answers (b) and (d). A current flowing counterclockwise in the outer loop of the figure produces a magnetic flux through the inner loop that is directed out of the page. If this current is increasing in time, the *change* in the flux is in the same direction as the flux itself (or out of the page). The induced current in the inner loop will then flow clockwise around the loop, producing a flux through the loop directed into the page, opposing the change in flux due to the increasing current in the outer loop. The flux through the inner loop is given by $\Phi_B = BA$, where B is the magnitude of the field and A is the area enclosed by the loop. The magnitude of the flux, and thus the magnitude of the rate of change of the flux, depends on the size of the area A.



ANS. FIG. OQ31.11

ANSWERS TO CONCEPTUAL QUESTIONS

CQ31.1 Recall that the net work done by a conservative force on an object is path independent; thus, if an object moves so that it starts and ends at the same place, the net conservative work done on it is zero. A positive electric charge carried around a circular electric field line in the direction of the field gains energy from the field every step of the way. It can be a test charge imagined to exist in vacuum or it can be an actual free charge participating in a current driven by an induced emf. By doing net work on an object carried around a closed path to its starting point, the magnetically-induced electric field exerts by definition a nonconservative force. We can get a larger and larger voltage just by looping a wire around into a coil with more and more turns.

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CQ31.2 The spacecraft is traveling through the magnetic field of the Earth. The magnetic flux through the coil must be changing to produce an emf, and thus a current. The orientation of the coil could be changing relative to the external magnetic field, or the field is changing through the coil because it is not uniform, or both.

CQ31.3 As water falls, it gains speed and kinetic energy. It then pushes against turbine blades, transferring its energy to the rotor coils of a large AC generator. The rotor of the generator turns within a strong magnetic field. Because the rotor is spinning, the magnetic flux through its coils changes in time as $\Phi_B = BA\cos\omega t$. Generated in the rotor is an induced emf of $\mathcal{E} = \frac{-Nd\Phi_B}{dt}$. This induced emf is the voltage driving the current in our electric power lines.

CQ31.4 Let us assume the north pole of the magnet faces the ring. As the bar magnet falls toward the conducting ring, a magnetic field is induced in the ring pointing upward. This upward directed field will oppose the motion of the magnet, preventing it from moving as a freely-falling body. Try it for yourself to show that an upward force also acts on the falling magnet if the south end faces the ring.

CQ31.5 To produce an emf, the magnetic flux through the loop must change. The flux cannot change if the orientation of the loop remains fixed in space because the magnetic field is uniform and constant. The flux does change if the loop is rotated so that the angle between the normal to the surface and the direction of the magnetic field changes.

CQ31.6 Yes. The induced eddy currents on the surface of the aluminum will slow the descent of the aluminum. In a strong field the piece may fall very slowly.

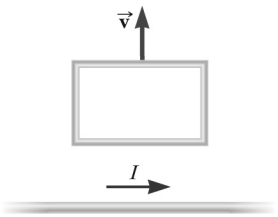
CQ31.7 Magnetic flux measures the "flow" of the magnetic field through a given area of a loop—even though the field does not actually flow. By changing the size of the loop, or the orientation of the loop and the field, one can change the magnetic flux through the loop, but the magnetic field will not change.

CQ31.8 The increasing counterclockwise current in the solenoid coil produces an upward magnetic field that increases rapidly. The increasing upward flux of this field through the ring induces an emf to produce clockwise current in the ring. The magnetic field of the solenoid has a radially outward component at each point on the ring. This field component exerts upward force on the current in the ring there. The whole ring feels a total upward force larger than its weight.

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CQ31.9 Oscillating current in the solenoid produces an always-changing magnetic field. Vertical flux through the ring, alternately increasing and decreasing, produces current in it with a direction that is alternately clockwise and counterclockwise. The current through the ring's resistance converts electrically transmitted energy into internal energy at the rate I^2R.

CQ31.10 (a) Counterclockwise. With the current in the long wire flowing in the direction shown in the figure, the magnetic flux through the rectangular loop is directed out of the page. As the loop moves away from the wire, the magnetic field through the loop becomes weaker, so the magnetic flux through the loop is decreasing in time, and the *change* in the flux is directed opposite to the flux itself (or into the page). The induced current will then flow counterclockwise around the loop, producing a flux directed out of the page through the loop and opposing the change in flux due to the decreasing flux through the loop.



ANS. FIG. CQ31.10

(b) Clockwise. In this case, as the loop moves toward from the wire, the magnetic field through the loop becomes stronger, so the magnetic flux through the loop is increasing in time, and the *change* in the flux has the same direction as the flux itself (or out of the page). The induced current will then flow clockwise around the loop, producing a flux directed into the page through the loop and opposing the change in flux due to the increasing flux through the loop.

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SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 31.1 Faraday's Law of Induction

***P31.1** From Equation 31.1, the induced emf is given by

$$|\mathcal{E}| = \left|\frac{\Delta\Phi_B}{\Delta t}\right| = \frac{\Delta\left(\vec{\mathbf{B}}\cdot\vec{\mathbf{A}}\right)}{\Delta t}$$

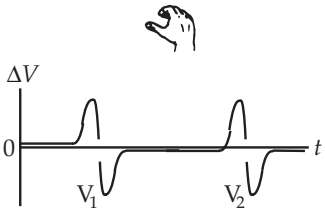
$$= \frac{(2.50\ \text{T} - 0.500\ \text{T})\left(8.00\times10^{-4}\ \text{m}^2\right)}{1.00\ \text{s}}\left(\frac{1\ \text{N}\cdot\text{s}}{1\ \text{T}\cdot\text{C}\cdot\text{m}}\right)\left(\frac{1\ \text{V}\cdot\text{C}}{1\ \text{N}\cdot\text{m}}\right)$$

$$= 1.60\ \text{mV}$$

We then find the current induced in the loop from

$$I_{\text{loop}} = \frac{\mathcal{E}}{R} = \frac{1.60\ \text{mV}}{2.00\ \Omega} = \boxed{0.800\ \text{mA}}$$

***P31.2** (a) Each coil has a pulse of voltage tending to produce counterclockwise current as the projectile approaches, and then a pulse of clockwise voltage as the projectile recedes.



ANS. FIG. P31.2

(b) $v = \dfrac{d}{t} = \dfrac{1.50\ \text{m}}{2.40\times10^{-3}\ \text{s}} = \boxed{625\ \text{m/s}}$

P31.3 (a) From Faraday's law,

$$\mathcal{E} = -N\frac{\Delta\Phi}{\Delta t} = -N\left(\frac{\Delta B}{\Delta t}\right)A\cos\theta$$

$$|\mathcal{E}| = \left|-(1)\left(\frac{B_f - B_i}{\Delta t}\right)\left(\pi r^2\right)\cos\theta\right|$$

$$= \left(\frac{1.50\ \text{T} - 0}{0.120\ \text{s}}\right)\left[\pi(0.001\ 60\ \text{m})^2\right](1)$$

$$= (12.5\ \text{T/s})\left[\pi(0.001\ 60\ \text{m})^2\right] = 1.01\times10^{-4}\ \text{T}$$

$$= \boxed{\begin{array}{l}101\ \mu\text{V tending to produce clockwise current}\\ \text{as seen from above}\end{array}}$$

(b) In case (a), the rate of change of the magnetic field was +12.5 T/s. In this case, the rate of change of the magnetic field is (–0.5 T – 1.5 T)/ 0.08 s = –25.0 T/s: it is twice as large in magnitude and in the opposite sense from the rate of change in case (a), so the emf is also

$\boxed{\text{twice as large in magnitude and in the opposite sense}}$.

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P31.4 From Equation 31.2,

$$\mathcal{E} = -N\frac{\Delta(BA\cos\theta)}{\Delta t} = -NB\pi r^2\left(\frac{\cos\theta_f - \cos\theta_i}{\Delta t}\right)$$

$$= -25.0\left(50.0\times10^{-6}\text{ T}\right)\left[\pi(0.500\text{ m})^2\right]\left(\frac{\cos 180° - \cos 0°}{0.200\text{ s}}\right)$$

$$\mathcal{E} = \boxed{+9.82\text{ mV}}$$

P31.5 With the field directed perpendicular to the plane of the coil, the flux through the coil is $\Phi_B = BA\cos 0° = BA$. For a single loop,

$$|\mathcal{E}| = \frac{\Delta\Phi_B}{\Delta t} = \frac{B(\Delta A)}{\Delta t}$$

$$= \frac{(0.150\text{ T})\left[\pi(0.120\text{ m})^2 - 0\right]}{0.200\text{ s}} = 3.39\times10^{-2}\text{ V} = \boxed{33.9\text{ mV}}$$

P31.6 With the field directed perpendicular to the plane of the coil, the flux through the coil is $\Phi_B = BA\cos 0° = BA$. As the magnitude of the field increases, the magnitude of the induced emf in the coil is

$$|\mathcal{E}| = \frac{\left|\Delta\Phi_B\right|}{\Delta t} = \left(\frac{\Delta B}{\Delta t}\right)A = \left(0.050\,0\text{ T/s}\right)\left[\pi(0.120\text{ m})^2\right]$$

$$= 2.26\times10^{-3}\text{ V} = \boxed{2.26\text{ mV}}$$

P31.7 The angle between the normal to the coil and the magnetic field is 90.0° – 28.0° = 62.0°. For a loop of N turns,

$$\mathcal{E} = -N\frac{d\Phi_B}{dt} = -N\frac{d}{dt}(BA\cos\theta)$$

$$\mathcal{E} = -NB\cos\theta\left(\frac{\Delta A}{\Delta t}\right)$$

$$= -200\left(50.0\times10^{-6}\text{ T}\right)(\cos 62.0°)\left(\frac{39.0\times10^{-4}\text{ m}^2}{1.80\text{ s}}\right)$$

$$= \boxed{-10.2\ \mu\text{V}}$$

P31.8 For a loop of N turns, the induced voltage is

$$\mathcal{E} = -N\frac{d\left(\vec{\mathbf{B}}\cdot\vec{\mathbf{A}}\right)}{dt} = -N\left(\frac{0 - B_iA\cos\theta}{\Delta t}\right)$$

$$= \frac{+200(1.60\text{ T})\left(0.200\text{ m}^2\right)\cos 0°}{20.0\times10^{-3}\text{ s}} = 3\ 200\text{ V}$$

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The induced current is then

$$I = \frac{\mathcal{E}}{R} = \frac{3\ 200\ \text{V}}{20.0\ \Omega} = \boxed{160\ \text{A}}$$

P31.9 Faradays law gives

$$|\mathcal{E}| = \left|\frac{\Delta\Phi_B}{\Delta t}\right| = N\left(\frac{dB}{dt}\right)A = N\left[\frac{d}{dt}\left(0.010\ 0t + 0.040\ 0t^2\right)\right]A$$

or $$|\mathcal{E}| = N\left(0.010\ 0 + 0.080\ 0t\right)A$$

where $|\mathcal{E}|$ is in volts, A is in meters squared, and t is in seconds. At $t = 5.00$ s, suppressing units,

$$|\mathcal{E}| = 30.0\left[0.010\ 0 + 0.080\ 0(5.00)\right]\left[\pi(0.040\ 0)^2\right]$$
$$= 6.18\times10^{-2} = \boxed{61.8\ \text{mV}}$$

P31.10 We have a stationary loop in an oscillating magnetic field that varies sinusoidally in time: $B = B_{\max}\sin\omega t,$ where $B_{\max} = 1.00\times10^{-8}$ T, $\omega = 2\pi f,$ and $f = 60.0$ Hz. The loop consists of a single band ($N = 1$) around the perimeter of a red blood cell with diameter $d = 8.00\times10^{-6}$ m and area $A = \pi d^2/4$. The induced emf is then

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -N\left(\frac{dB}{dt}\right)A$$
$$= -N\frac{d}{dt}\left(B_{\max}\sin\omega t\right)A = -\omega NAB_{\max}\cos\omega t$$

Comparing this expression to $\mathcal{E} = \mathcal{E}_{\max}\cos\omega t,$ we see that $\mathcal{E}_{\max} = \omega NAB_{\max}$. Therefore,

$$\mathcal{E}_{\max} = \omega NAB_{\max}$$
$$= \left[2\pi(60.0\ \text{Hz})\right](1)\left[\frac{\pi\left(8.00\times10^{-6}\ \text{m}\right)^2}{4}\right]\left(1.00\times10^{-3}\ \text{T}\right)$$
$$= \boxed{1.89\times10^{-11}\ \text{V}}$$

P31.11 The symbol for the radius of the ring is r_1, and we use R to represent its resistance. The emf induced in the ring is

$$\mathcal{E} = -\frac{d}{dt}(BA\cos\theta) = -\frac{d}{dt}(0.500\mu_0 nIA\cos0°) = -0.500\mu_0 nA\frac{dI}{dt}$$

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Note that A must be interpreted as the area $A = \pi r_2^2$ of the solenoid, where the field is strong:

$$\mathcal{E} = -0.500(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A})(1\,000\ \text{turns/m}) \times [\pi(0.030\ 0\ \text{m})^2](270\ \text{A/s})$$

$$\mathcal{E} = \left(-4.80 \times 10^{-4}\,\frac{\text{T}\cdot\text{m}^2}{\text{s}}\right)\left(\frac{1\ \text{N}\cdot\text{s}}{\text{C}\cdot\text{m}\cdot\text{T}}\right)\left(\frac{1\ \text{V}\cdot\text{C}}{\text{N}\cdot\text{m}}\right) = -4.80\times 10^{-4}\ \text{V}$$

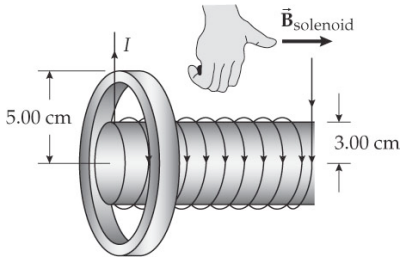
(a) The negative sign means that the current in the ring is counterclockwise, opposite to the current in the solenoid. Its magnitude is

$$I_{\text{ring}} = \frac{|\mathcal{E}|}{R} = \frac{0.000\,480\ \text{V}}{0.000\,300\ \Omega} = \boxed{1.60\ \text{A}}$$

(b) $$B_{\text{ring}} = \frac{\mu_0 I}{2r_1} = \frac{\left(4\pi \times 10^{-7}\ \text{T}\cdot\text{m/A}\right)(1.60\ \text{A})}{2(0.050\ 0\ \text{m})}$$

$$= 2.01\times 10^{-5}\ \text{T} = \boxed{20.1\ \mu\text{T}}$$

(c) The solenoid's field points to the right through the ring, and is increasing, so to oppose the increasing field, B_{ring} points to the **left**.



ANS. FIG. P31.11

P31.12 See ANS. FIG. P31.11. The emf induced in the ring is

$$|\mathcal{E}| = \frac{d(BA)}{dt} = \frac{1}{2}\frac{d}{dt}(\mu_0 nI)A = \frac{1}{2}\mu_0 n\frac{dI}{dt}\pi r_2^2 = \frac{1}{2}\mu_0 n\pi r_2^2\frac{\Delta I}{\Delta t}$$

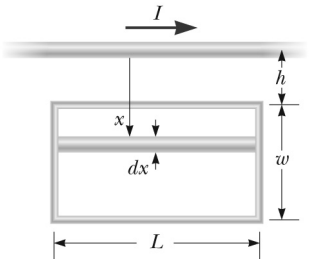
(a) $$I_{\text{ring}} = \frac{\mathcal{E}}{R} = \boxed{\frac{\mu_0 n\pi r_2^2}{2R}\frac{\Delta I}{\Delta t}}$$, counterclockwise as viewed from the left end.

(b) $$B = \frac{\mu_0 I}{2r_1} = \boxed{\frac{\mu_0^2 n\pi r_2^2}{4r_1 R}\frac{\Delta I}{\Delta t}}$$

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(c) The solenoid's field points to the right through the ring, and is increasing, so to oppose the increasing field, B_{ring} points to the left.

P31.13 (a) At a distance x from the long, straight wire, the magnetic field is $B = \dfrac{\mu_0 I}{2\pi x}$. The flux through a small rectangular element of length L and width dx within the loop is



ANS. FIG. P31.13

$$d\Phi_B = \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = \frac{\mu_0 I}{2\pi x} L dx:$$

$$\Phi_B = \int_h^{h+w} \frac{\mu_0 IL}{2\pi} \frac{dx}{x} = \boxed{\frac{\mu_0 IL}{2\pi} \ln\left(\frac{h+w}{h}\right)}$$

(b) $$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\left[\frac{\mu_0 IL}{2\pi} \ln\left(\frac{h+w}{h}\right)\right] = -\left[\frac{\mu_0 L}{2\pi} \ln\left(\frac{h+w}{h}\right)\right]\frac{dI}{dt}$$

where $\dfrac{dI}{dt} = \dfrac{d}{dt}(a+bt) = b$:

$$\mathcal{E} = -\frac{\left(4\pi \times 10^{-7}\ \text{T} \cdot \text{m/A}\right)(1.00\ \text{m})}{2\pi} \times \ln\left(\frac{0.010\ 0\ \text{m} + 0.100\ \text{m}}{0.010\ 0\ \text{m}}\right)(10.0\ \text{A/s})$$

$$= -4.80 \times 10^{-6}\ \text{V}$$

Therefore, the emf induced in the loop is $\boxed{4.80\ \mu\text{V}}$.

(c) The long, straight wire produces magnetic flux into the page through the rectangle, shown in ANS. FIG. P31.13. As the magnetic flux increases, the rectangle produces its own magnetic field out of the page to oppose the increase in flux. The induced current creates this opposing field by traveling counterclockwise around the loop.

P31.14 The magnetic field lines are confined to the interior of the solenoid, so even though the coil has a larger area, the flux through the coil is the same as the flux through the solenoid:

$$\Phi_B = \left(\mu_0 nI\right) A_{\text{solenoid}}$$

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$$\mathcal{E} = -N\frac{d\Phi_B}{dt} = -N\mu_0 n\left(\pi r_{\text{solenoid}}^2\right)\frac{dI}{dt}$$
$$= -(15)\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(1.00\times10^{3}\ \text{m}^{-1}\right)$$
$$\times\pi(0.020\ 0\ \text{m})^2(600)\cos(120t)$$
$$= -0.014\ 2\cos(120t)$$

$$\boxed{\mathcal{E} = -\left(1.42\times10^{-2}\right)\cos(120t),\ \text{where } t \text{ is in seconds and } \mathcal{E} \text{ is in V.}}$$

P31.15 The initial magnetic field inside the solenoid is

$$B = \mu_0 nI = \left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(\frac{100}{0.200\ \text{m}}\right)(3.00\ \text{A})$$
$$= 1.88\times10^{-3}\ \text{T}$$

(a) $$\Phi_B = BA\cos\theta = \left(1.88\times10^{-3}\ \text{T}\right)\left(1.00\times10^{-2}\ \text{m}\right)^2\cos 0°$$

$$= \boxed{1.88\times10^{-7}\ \text{T}\cdot\text{m}^2}$$

(b) When the current is zero, the flux through the loop is $\Phi_B = 0$ and the average induced emf has been

$$|\mathcal{E}| = \frac{|\Delta\Phi_B|}{\Delta t} = \frac{\left|0 - 1.88\times10^{-7}\ \text{T}\cdot\text{m}^2\right|}{3.00\ \text{s}} = \boxed{6.28\times10^{-8}\ \text{V}}$$

P31.16 The solenoid creates a magnetic field

$$B = \mu_0 nI = (4\pi\times10^{-7}\ \text{N/A}^2)(400\ \text{turns/m})(30.0\ \text{A})(1 - e^{-1.60\,t})$$

$$B = (1.51\times10^{-2}\ \text{N/m}\cdot\text{A})(1 - e^{-1.60\,t})$$

The magnetic flux through one turn of the flat coil is $\Phi_B = \int B\,dA\cos\theta$, but since $dA\cos\theta$ refers to the area perpendicular to the flux, and the magnetic field is uniform over the area A of the flat coil, this integral simplifies to

$$\Phi_B = B\int dA = B\left(\pi R^2\right)$$
$$= \left(1.51\times10^{-2}\ \text{N/m}\cdot\text{A}\right)\left(1 - e^{-1.60t}\right)\left[\pi(0.060\ 0\ \text{m})^2\right]$$
$$= \left(1.71\times10^{-4}\ \text{N/m}\cdot\text{A}\right)\left(1 - e^{-1.60t}\right)$$

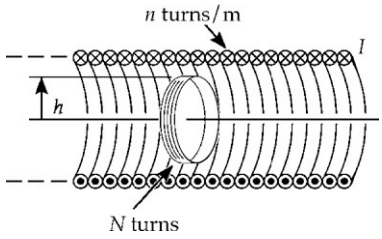
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The emf generated in the N-turn coil is $\mathcal{E} = -N\, d\Phi_B/dt$. Because t has the standard unit of seconds, the factor 1.60 must have the unit s^{-1}.

$$\mathcal{E} = -(250)\left(1.71\times10^{-4}\,\frac{\text{N}\cdot\text{m}}{\text{A}}\right)\frac{d\left(1-e^{-1.60\,t}\right)}{dt}$$

$$= -\left(0.042\,6\,\frac{\text{N}\cdot\text{m}}{\text{A}}\right)(1.60\ \text{s}^{-1})e^{t-1.60}$$

$$\boxed{\mathcal{E} = 68.2e^{-1.60t},\ \text{where } t \text{ is in seconds and } \mathcal{E} \text{ is in mV.}}$$



ANS. FIG. P31.16

P31.17 Faraday's law, $\mathcal{E} = -N\dfrac{d\Phi_B}{dt}$, becomes here

$$\mathcal{E} = -N\frac{d}{dt}\left(BA\cos\theta\right) = -NA\cos\theta\frac{dB}{dt}$$

The magnitude of the emf is

$$|\mathcal{E}| = NA\cos\theta\left(\frac{\Delta B}{\Delta t}\right)$$

The area is

$$A = \frac{|\mathcal{E}|}{N\cos\theta\left(\dfrac{\Delta B}{\Delta t}\right)}$$

$$A = \frac{80.0\times10^{-3}\ \text{V}}{50\left(\cos30.0^{\circ}\right)\left(\dfrac{600\times10^{-6}\ \text{T} - 200\times10^{-6}\ \text{T}}{0.400\ \text{s}}\right)} = 1.85\ \text{m}^2$$

Each side of the coil has length $d = \sqrt{A}$, so the total length of the wire is

$$L = N(4d) = 4N\sqrt{A} = (4)(50)\sqrt{1.85\ \text{m}^2} = \boxed{272\ \text{m}}$$

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P31.18 (a) Suppose, first, that the central wire is long and straight. The enclosed current of unknown amplitude creates a circular magnetic field around it, with the magnitude of the field given by Ampère's law.

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = \mu_0 I: \qquad B = \frac{\mu_0 I_{\max} \sin \omega t}{2\pi R}$$

at the location of the Rogowski coil, which we assume is centered on the wire. This field passes perpendicularly through each turn of the toroid, producing flux

$$\vec{\mathbf{B}} \cdot \vec{\mathbf{A}} = \frac{\mu_0 I_{\max} A}{2\pi R} \sin \omega t$$

The toroid has $2\pi Rn$ turns. As the magnetic field varies, the emf induced in it is

$$\mathcal{E} = -N \frac{d}{dt} \vec{\mathbf{B}} \cdot \vec{\mathbf{A}} = -2\pi Rn \frac{\mu_0 I_{\max} A}{2\pi R} \frac{d}{dt} \sin \omega t$$
$$= -\mu_0 I_{\max} nA\omega \cos \omega t$$

This is an alternating voltage with amplitude $\mathcal{E}_{\max} = \mu_0 nA\omega I_{\max}$. Measuring the amplitude determines the size $I_{\max}$ of the central current. Our assumptions that the central wire is long and straight and passes perpendicularly through the center of the Rogowski coil are all unnecessary.

(b) If the wire is not centered, the coil will respond to stronger magnetic fields on one side, but to correspondingly weaker fields on the opposite side. The emf induced in the coil is proportional to the line integral of the magnetic field around the circular axis of the toroid. Ampère's law says that this line integral depends only on the amount of current the coil encloses. It does not depend on the shape or location of the current within the coil, or on any currents outside the coil.

P31.17 In a toroid, all the flux is confined to the inside of the toroid. From Equation 30.16, the field inside the toroid at a distance r from its center is

$$B = \frac{\mu_0 NI}{2\pi r}$$

The magnetic flux is then

$$\Phi_B = \int B\, dA = \frac{\mu_0 NI_{\max}}{2\pi} \sin \omega t \int \frac{a\, dr}{r}$$
$$= \frac{\mu_0 NI_{\max}}{2\pi} a \sin \omega t \ln\left(\frac{b+R}{R}\right)$$

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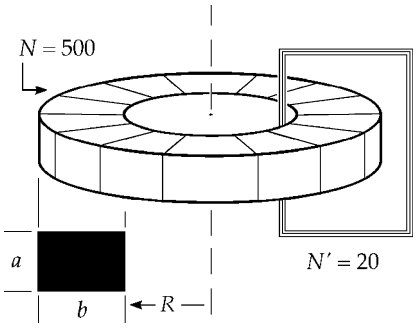
and the induced emf is

$$\mathcal{E} = N' \frac{d\Phi_B}{dt} = N' \left(\frac{\mu_0 N I_{\max}}{2\pi} \right) \omega a \ln \left(\frac{b+R}{R} \right) \cos \omega t$$

Substituting numerical values and suppressing units,

$$\mathcal{E} = 20 \frac{\left(4\pi \times 10^{-7}\right)(500)(50.0)}{2\pi} \times \left[2\pi(60.0)\right](0.020\,0) \ln \left(\frac{0.030\,0 + 0.040\,0}{0.040\,0} \right) \cos \omega t$$

$$\boxed{\mathcal{E} = 0.422 \cos \omega t \text{ where } \mathcal{E} \text{ is in volts and } t \text{ is in seconds.}}$$



ANS. FIG. P31.19

P31.20 In Figure P31.20, the original magnetic field points into the page and is increasing. The induced emf in the upper loop attempts to generate a counterclockwise current in order to produce a magnetic field out of the page that opposes the increasing external magnetic flux. The induced emf in the lower loop also must attempt to generate a counterclockwise current in order to produce a magnetic field out of the page that opposes the increasing external magnetic flux. Because of the crossing over between the two loops, the emf generated in the loops will be in opposite directions. Therefore, the magnitude of the net emf generated is

$$\mathcal{E}_{\text{net}} = \mathcal{E}_2 - \mathcal{E}_1 = A_2 \frac{dB}{dt} - A_1 \frac{dB}{dt} = \left(\pi r_2^2 - \pi r_1^2\right) \frac{dB}{dt}$$

$$= \pi \frac{dB}{dt} \left(r_2^2 - r_1^2\right)$$

where the upper loop is loop 1 and the lower one is loop 2.

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(a) The induced current will be the ratio of the net emf to the total resistance of the loops:

$$I = \frac{\mathcal{E}_{\text{net}}}{R} = \frac{\pi \frac{dB}{dt}\left(r_2^2 - r_1^2\right)}{\left(\frac{R}{\ell}\right)\ell_{\text{total}}} = \frac{\pi \frac{dB}{dt}\left(r_2^2 - r_1^2\right)}{\left(\frac{R}{\ell}\right)\left(2\pi r_2 + 2\pi r_1\right)}$$

$$= \frac{\frac{dB}{dt}\left(r_2^2 - r_1^2\right)}{2\left(\frac{R}{\ell}\right)\left(r_2 + r_1\right)} = \frac{\frac{dB}{dt}\left(r_2 - r_1\right)\left(r_2 + r_1\right)}{2\left(\frac{R}{\ell}\right)\left(r_2 + r_1\right)}$$

$$= \frac{\frac{dB}{dt}\left(r_2 - r_1\right)}{2\left(\frac{R}{\ell}\right)}$$

Substitute numerical values:

$$I = \frac{(2.00\ \text{T/s})(0.090\ 0\ \text{m} - 0.050\ 0\ \text{m})}{2(3.00\ \Omega/\text{m})} = \boxed{0.013\ 3\ \text{A}}$$

(b) The emf in each loop is trying to push charge in opposite directions through the wire, but the emf in the lower loop is larger because its area is larger (changing flux is proportional to the area of the loop), so the lower loop "wins": the current is counterclockwise in the lower loop and clockwise in the upper loop.

Section 31.2 Motional emf

Section 31.3 Lenz's Law

***P31.21** The angular speed of the rotor blades is

$$\omega = (2.00\ \text{rev/s})(2\pi\ \text{rad/rev}) = 4.00\pi\ \text{rad/s}$$

Thus, the motional emf is then

$$\mathcal{E} = \frac{1}{2}B\omega\ell^2 = \frac{1}{2}\left(50.0\times10^{-6}\ \text{T}\right)(4.00\pi\ \text{rad/s})(3.00\ \text{m})^2$$

$$= \boxed{2.83\ \text{mV}}$$

P31.22 (a) $\vec{\mathbf{B}}_{\text{ext}} = B_{\text{ext}}\hat{\mathbf{i}}$ and B_{ext} decreases; therefore, the induced field is $\vec{\mathbf{B}}_{\text{induced}} = B_{\text{induced}}\hat{\mathbf{i}}$ (to the right) and the current in the resistor is directed from a to b, to the right.

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(b) $\vec{\mathbf{B}}_{\text{ext}} = B_{\text{ext}}\left(-\hat{\mathbf{i}}\right)$ increases; therefore, the induced field $\vec{\mathbf{B}}_{\text{induced}} = B_{\text{induced}}\left(+\hat{\mathbf{i}}\right)$ is to the right, and the current in the resistor is directed from *a* to *b*, out of the page in the textbook picture.

(c) $\vec{\mathbf{B}}_{\text{ext}} = B_{\text{ext}}\left(-\hat{\mathbf{k}}\right)$ into the paper and B_{ext} decreases; therefore, the induced field is $\vec{\mathbf{B}}_{\text{induced}} = B_{\text{induced}}\left(-\hat{\mathbf{k}}\right)$ into the paper, and the current in the resistor is directed from *a* to *b*, to the right.

P31.23 The motional emf induced in a conductor is proportional to the component of the magnetic field perpendicular to the conductor and to its velocity.

$$\mathcal{E} = B\ell v = \left(35.0\times10^{-6}\ \text{T}\right)(15.0\ \text{m})(25.0\ \text{m/s})$$
$$= 1.31\times10^{-2}\ \text{V} = \boxed{13.1\ \text{mV}}$$

P31.24 (a) The potential difference is equal to the motional emf and is given by

$$\mathcal{E} = B\ell v = \left(1.20\times10^{-6}\ \text{T}\right)(14.0\ \text{m})(70.0\ \text{m/s})$$
$$= 1.18\times10^{-3}\ \text{V} = \boxed{11.8\ \text{mV}}$$

(b) A free positive test charge in the wing feels a magnetic force in direction $\vec{\mathbf{v}}\times\vec{\mathbf{B}}$ = (north) × (down) = (west): it migrates west. The wingtip on the pilot's left is positive.

(c) No change. A positive test charge in the wing feels a magnetic force in direction $\vec{\mathbf{v}}\times\vec{\mathbf{B}}$ = (east) × (down) = (north): it migrates north. The left wingtip is north of the pilot.

(d) No. If you try to connect the wings to a circuit containing the light bulb, you must run an extra insulated wire along the wing. In a uniform field the total emf generated in the one-turn coil is zero.

P31.25 (a) The motional emf induced in a conductor is proportional to the component of the magnetic field perpendicular to the conductor and to its velocity; in this case, the vertical component of the Earth's magnetic field is perpendicular to both. Thus, the magnitude of the motional emf induced in the wire is

$$\mathcal{E} = B_{\perp}\ell v = \left[\left(50.0\times10^{-6}\ \text{T}\right)\sin 53.0^{\circ}\right](2.00\ \text{m})(0.500\ \text{m/s})$$
$$= 3.99\times10^{-5}\ \text{V} = \boxed{39.9\ \mu\text{V}}$$

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(b) Imagine holding your right hand horizontal with the fingers pointing north (the direction of the wire's velocity), such that when you close your hand the fingers curl downward (in the direction of $B_\perp$). Your thumb will then be pointing westward. By the right-hand rule, the magnetic force on charges in the wire would tend to move positive charges westward.

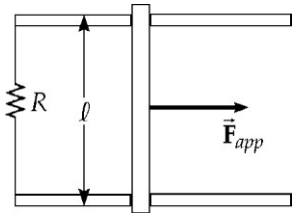
The west end is positive.

***P31.26** See ANS. FIG. P31.26. The current is given by

$$I = \frac{\mathcal{E}}{R} = \frac{B\ell v}{R}$$

Solving for the velocity gives

$$v = \frac{IR}{B\ell} = \frac{(0.500\text{ A})(6.00\ \Omega)}{(2.50\text{ T})(1.20\text{ m})} = \boxed{1.00\text{ m/s}}$$



ANS. FIG. P31.26

P31.27 (a) Refer to ANS. FIG. P31.26 above. At constant speed, the net force on the moving bar equals zero, or

$$\left|\vec{\mathbf{F}}_{\mathbf{app}}\right| = I\left|\vec{\mathbf{L}} \times \vec{\mathbf{B}}\right|$$

where the current in the bar is $I = \mathcal{E}/R$ and the motional emf is $\mathcal{E} = B\,\ell v$. Therefore,

$$F_B = \frac{B\ell v}{R}(\ell B) = \frac{B^2\ell^2 v}{R} = \frac{(2.50\text{ T})^2(1.20\text{ m})^2(2.00\text{ m/s})}{6.00\ \Omega}$$
$$= 3.00\text{ N}$$

The applied force is $\boxed{3.00\text{ N to the right}}$.

(b) $P = I^2R = \dfrac{B^2\ell^2v^2}{R} = 6.00\text{ W}$ or $P = Fv = \boxed{6.00\text{ W}}$

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***P31.28** With v representing the initial speed of the bar, let u represent its speed at any later time. The motional emf induced in the bar is $\mathcal{E} = B\ell u$. The induced current is $I = \dfrac{\mathcal{E}}{R} = \dfrac{B\ell u}{R}$. The magnetic force on the bar is backward $F = -I\ell B = -\dfrac{B^2\ell^2 u}{R} = \dfrac{mdu}{dt}$.

Method one: To find u as a function of time, we separate variables thus:

$$-\frac{B^2\ell^2}{Rm}dt = \frac{du}{u}$$

$$\int_0^t -\frac{B^2\ell^2}{Rm}dt = \int_v^u \frac{du}{u}$$

$$-\frac{B^2\ell^2}{Rm}(t-0) = \ln u - \ln v = \ln\frac{u}{v}$$

$$e^{-B^2\ell^2 t/Rm} = \frac{u}{v}$$

$$u = ve^{-B^2\ell^2 t/Rm} = \frac{dx}{dt}$$

The distance traveled is given by

$$\int_0^{x_{\max}} dx = \int_0^{\infty} ve^{-B^2\ell^2 t/Rm}\,dt = v\left(-\frac{Rm}{B^2\ell^2}\right)\int_0^{\infty} e^{-B^2\ell^2 t/Rm}\left(-\frac{-B^2\ell^2 dt}{Rm}\right)$$

$$x_{\max} - 0 = -\frac{Rmv}{B^2\ell^2}\left[e^{-\infty} - e^{-0}\right] = \boxed{\frac{Rmv}{B^2\ell^2}}$$

Method two: Newton's second law is

$$-\frac{B^2\ell^2 u}{R} = -\frac{B^2\ell^2}{R}\frac{dx}{dt} = m\frac{du}{dt}$$

$$mdu = -\frac{B^2\ell^2}{R}dx$$

Direct integration from the initial to the stopping point gives

$$\int_v^0 m\,du = \int_0^{x_{\max}} -\frac{B^2\ell^2}{R}dx$$

$$m(0-v) = -\frac{B^2\ell^2}{R}(x_{\max} - 0)$$

$$x_{\max} = \frac{mvR}{B^2\ell^2}$$

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***P31.29** The magnetic force on the rod is given by

$$F_B = I\ell B$$

and the motional emf by

$$\mathcal{E} = B\ell v$$

The current is given by $I = \dfrac{\mathcal{E}}{R} = \dfrac{B\ell v}{R}$, so $B = \dfrac{IR}{\ell v}$.

(a) $F_B = \dfrac{I^2 \ell R}{\ell v}$ and $I = \sqrt{\dfrac{F_B v}{R}} = \sqrt{\dfrac{(1.00\text{ N})(2.00\text{ m/s})}{8.00\ \Omega}} = \boxed{0.500\text{ A}}$

(b) The rate at which energy is delivered to the resistor is the power delivered, given by

$$P = I^2 R = (0.500\text{ A})^2 (8.00\ \Omega) = \boxed{2.00\text{ W}}$$

(c) For constant force, $P = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}} = (1.00\text{ N})(2.00\text{ m/s}) = \boxed{2.00\text{ W}}$.

P31.30 To maximize the motional emf, the automobile must be moving east or west. Only the component of the magnetic field to the north generates an emf in the moving antenna. Therefore, the maximum motional emf is

$$\mathcal{E}_{\max} = B\ell v \cos\theta$$

Let's solve for the unknown speed of the car:

$$v = \frac{\mathcal{E}_{\max}}{B\ell \cos\theta}$$

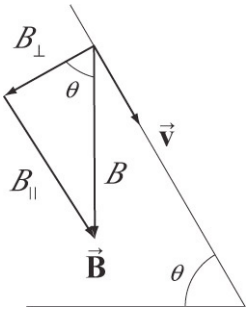
Substitute numerical values:

$$v = \frac{4.50 \times 10^{-3}\text{ V}}{(50.0 \times 10^{-6}\text{ T})(1.20\text{ m})\cos 65.0^\circ} = 177\text{ m/s}$$

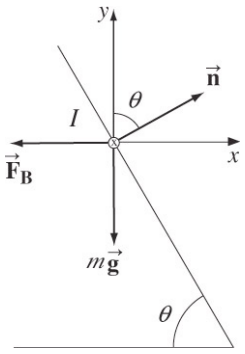
This is equivalent to about 640 km/h or 400 mi/h, much faster than the car could drive on the curvy road and much faster than any standard automobile could drive in general.

P31.31 The motional emf induced in a conductor is proportional to the component of the magnetic field perpendicular to the conductor and to its velocity. The total field is perpendicular to the conductor, but not to its velocity. As shown in the left figure, the component of the field perpendicular to the velocity is $B_\perp = B\cos\theta$. The motion of the bar down the rails produces an induced emf $\mathcal{E} = B_\perp \ell v = B\ell v \cos\theta$ that pushes charge into the page. The induced emf produces a current $I = \mathcal{E}/R = B\ell v \cos\theta / R$, where we assume that significant resistance is present only in the resistor. Because current in the bar travels into the page, and the field is downward, a magnetic force acts on the bar to the left: its magnitude is $F = I\ell B \sin 90.0^\circ = I\ell B = B^2 \ell^2 v \cos\theta / R$.

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ANS. FIG. P31.31(a)



ANS. FIG. P31.31(b)

In the free-body diagram shown in ANS. FIG. P31.31(b), it is convenient to use a coordinate system with axes vertical and horizontal. The force relationships are

$$\sum F_x = -F + n\sin\theta = 0 \quad \rightarrow \quad n\sin\theta = F = B^2\ell^2 v\cos\theta/R$$

$$\sum F_y = -mg + n\cos\theta = 0 \quad \rightarrow \quad n\cos\theta = mg$$

Dividing the first by the second equation, we get

$$\frac{\cancel{n}\sin\theta}{\cancel{n}\cos\theta} = \frac{B^2\ell^2 v\cos\theta/R}{mg} \quad \rightarrow \quad v = \frac{mgR\sin\theta}{B^2\ell^2\cos^2\theta}$$

Substituting numerical values,

$$v = \frac{(0.200\text{ kg})(9.80\text{ m/s}^2)(1.00\ \Omega)\sin 25.0^\circ}{(0.500\text{ T})^2(1.20\text{ m})^2\cos^2 25.0^\circ} = \boxed{2.80\text{ m/s}}$$

P31.32 Refer to ANS. FIG. P31.31 above. The motional emf induced in a conductor is proportional to the component of the magnetic field perpendicular to the conductor and to its velocity. The total field is perpendicular to the conductor, but not to its velocity. As shown in the left figure, the component of the field perpendicular to the velocity is $B_\perp = B\cos\theta$. The motion of the bar down the rails produces an induced emf $\mathcal{E} = B_\perp \ell v = B\ell v\cos\theta$ that pushes charge into the page. The induced emf produces a current $I = \mathcal{E}/R = B\ell v\cos\theta/R$, where we assume that significant resistance is present only in the resistor. Because current in the bar travels into the page, and the field is downward, a magnetic force acts on the bar to the left: its magnitude is $F = I\ell B\sin 90.0^\circ = I\ell B = B^2\ell^2 v\cos\theta/R$. In the free-body diagram shown in ANS. FIG. P31.31(b), it is convenient to use a coordinate system with axes vertical and horizontal. The force relationships are

$$\sum F_x = -F + n\sin\theta = 0 \quad \rightarrow \quad n\sin\theta = F = B^2\ell^2 v\cos\theta/R$$

$$\sum F_y = -mg + n\cos\theta = 0 \quad \rightarrow \quad n\cos\theta = mg$$

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Dividing the first by the second equation, we get

$$\frac{\cancel{m}\sin\theta}{\cancel{m}\cos\theta} = \frac{B^2\ell^2 v\cos\theta/R}{mg} \quad \rightarrow \quad v = \boxed{\frac{mgR\sin\theta}{B^2\ell^2\cos^2\theta}}$$

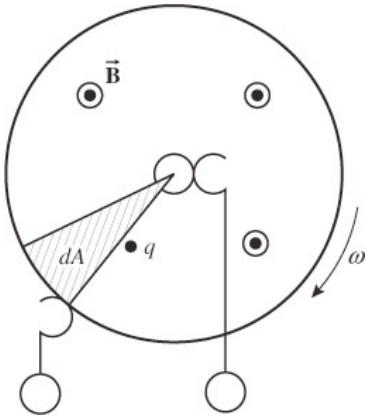
P31.33 From Example 31.4, the magnitude of the emf is

$$|\mathcal{E}| = B\left(\frac{1}{2}r^2\omega\right)$$

$$= (0.9\ \text{N}\cdot\text{s/C}\cdot\text{m})\left[\frac{1}{2}(0.4\ \text{m})^2(3\,200\ \text{rev/min})\right]\left(\frac{2\pi\ \text{rad/rev}}{60\ \text{s/min}}\right)$$

$$|\mathcal{E}| = \boxed{24.1\ \text{V}}$$

A free positive charge q, represented in our version of the diagram, turning with the disk, feels a magnetic force $q\vec{\mathbf{v}}\times\vec{\mathbf{B}}$  radially inward. Thus the outer contact is $\boxed{\text{negative}}$.



ANS. FIG. P31.33

P31.34 (a) The motional emf induced in the bar must be $\mathcal{E} = IR$, where I is the current in this series circuit. Since $\mathcal{E} = B\ell v$, the speed of the moving bar must be

$$v = \frac{\mathcal{E}}{B\ell} = \frac{IR}{B\ell} = \frac{(8.50\times10^{-3}\ \text{A})(9.00\ \Omega)}{(0.300\ \text{T})(0.350\ \text{m})} = \boxed{0.729\ \text{m/s}}$$

(b) The flux through the closed loop formed by the rails, the bar, and the resistor is directed into the page and is increasing in magnitude. To oppose this change in flux, the current must flow in a manner so as to produce flux out of the page through the area enclosed by the loop. This means the current will flow $\boxed{\text{counterclockwise}}$.

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(c) The rate at which energy is delivered to the resistor is

$$P = I^2R = \left(8.50\times10^{-3}\ \text{A}\right)^2(9.00\ \Omega)$$
$$= 6.50\times10^{-4}\ \text{W} = \boxed{0.650\ \text{mW}}$$

(d) Work is being done by the external force, which is transformed into internal energy in the resistor.

P31.35 The speed of waves on the wire is

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{mg}{\mu}} = \sqrt{\frac{267\ \text{N}}{3.00\times10^{-3}\ \text{kg/m}}} = 298\ \text{m/s}$$

In the simplest standing-wave vibration state,

$$d_{NN} = 0.64\ \text{m} = \frac{\lambda}{2} \rightarrow \lambda = 1.28\ \text{m}$$

and $$f = \frac{v}{\lambda} = \frac{298\ \text{m/s}}{1.28\ \text{m}} = 233\ \text{Hz}$$

(a) The changing flux of magnetic field through the circuit containing the wire will drive current to the left in the wire as it moves up and to the right as it moves down. The emf will have this same frequency of $\boxed{233\ \text{Hz}}$.

(b) The vertical coordinate of the center of the wire is described by

$$x = A\cos\omega t = A\cos 2\pi ft$$

Its velocity is $v = \frac{dx}{dt} = -2\pi fA\sin 2\pi ft$.

Its maximum speed is $v_{\max} = 2\pi fA$.

The induced emf is $\mathcal{E} = -B\ell v$, with amplitude

$$\mathcal{E}_{\max} = B\ell v_{\max} = B\ell 2\pi fA$$
$$= \left(4.50\times10^{-3}\ \text{T}\right)(0.0200\ \text{m})2\pi(233\ \text{Hz})(0.015\,0\ \text{m})$$
$$= 1.98\times10^{-3}\ \text{V} = \boxed{1.98\ \text{mV}}$$

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P31.36 (a) The force on the side of the coil entering the field (consisting of N wires) is

$$F = N(ILB) = N(IwB)$$

The induced emf in the coil is

$$|\mathcal{E}| = N\frac{d\Phi_B}{dt} = N\frac{d(Bwx)}{dt} = NBwv$$

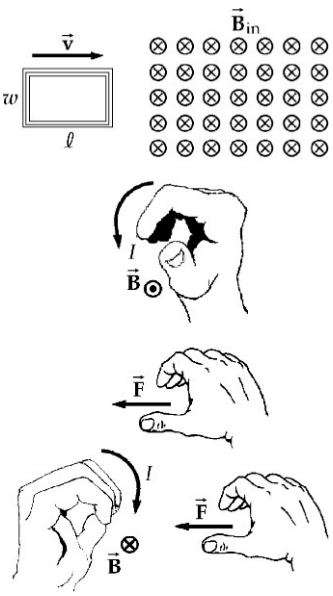
so the current is $I = \dfrac{|\mathcal{E}|}{R} = \dfrac{NBwv}{R}$

counterclockwise.

The force on the leading side of the coil is then:

$$F = N\left(\frac{NBwv}{R}\right)wB$$

$$= \boxed{\frac{N^2B^2w^2v}{R} \text{ to the left}}$$



ANS. FIG. P31.36

(b) Once the coil is entirely inside the field,

$$\Phi_B = NBA = \text{constant}$$

so $\mathcal{E} = 0,\ I = 0,$ and $F = \boxed{0}$.

(c) As the coil starts to leave the field, the flux *decreases* at the rate Bwv, so the magnitude of the current is the same as in part (a), but now the current is clockwise. Thus, the force exerted on the trailing side of the coil is:

$$F = \boxed{\frac{N^2B^2w^2v}{R} \text{ to the left again}}$$

P31.37 The emfs induced in the rods are proportional to the lengths of the sections of the rods between the rails. The emfs are $\mathcal{E}_1 = B\ell v_1$ with positive end downward, and $\mathcal{E}_2 = B\ell v_2$ with positive end upward, where $\ell = d = 10.0$ cm is the distance between the rails.

We apply Kirchhoff's laws. We assume current I_1 travels downward in the left rod, current I_2 travels upward in the right rod, and current I_3 travels upward in the resisitor R_3.

For the left loop, $+B\ell v_1 - I_1R_1 - I_3R_3 = 0$ **[1]**

For the right loop, $+B\ell v_2 - I_2R_2 + I_3R_3 = 0$ **[2]**

At the top junction, $I_1 = I_2 + I_3$ **[3]**

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Substituting [3] into [1] gives

$$B\ell v_1 - I_1R_1 - I_3R_3 = 0$$

$$B\ell v_1 - (I_2 + I_3)R_1 - I_3R_3 = 0$$

$$I_2R_1 + I_3(R_1 + R_3) = B\ell v_1 \qquad \textbf{[4]}$$

Now using [2] and [4] to solve for I_2,

$$I_2 = \frac{B\ell v_2 + I_3R_3}{R_2} = \frac{B\ell v_1 - I_3(R_1 + R_3)}{R_1}$$

then equating gives

$$(B\ell v_2 + I_3R_3)R_1 = \left[B\ell v_1 - I_3(R_1 + R_3)\right]R_2$$

$$I_3\left[R_3R_1 + (R_1 + R_3)R_2\right] = B\ell v_1R_2 - B\ell v_2R_1$$

Solving for I3 gives

$$I_3 = B\ell\frac{(v_1R_2 - v_2R_1)}{R_1R_2 + R_1R_3 + R_2R_3}$$

Substituting numerical values, and noting that

$$\begin{aligned} R_1R_2 + R_1R_3 + R_2R_3 &= (10.0\ \Omega)(15.0\ \Omega) \\ &\quad + (10.0\ \Omega)(5.00\ \Omega) + (15.0\ \Omega)(5.00\ \Omega) \\ &= 275\ \Omega^2 \end{aligned}$$

we obtain

$$\begin{aligned} I_3 &= (0.010\ 0\ \text{T})(0.100\ \text{m}) \\ &\quad \times \frac{\left[(4.00\ \text{m/s})(15.0\ \Omega) - (2.00\ \text{m/s})(10.0\ \Omega)\right]}{275\ \Omega^2} \\ &= 1.45 \times 10^{-4}\ \text{A} \end{aligned}$$

Therefore, $I_3 =$ $\boxed{145\ \mu\text{A upward in the picture}}$, as was originally chosen.

P31.38 (a) The induced emf is $\mathcal{E} = B\ell v$, where B is the magnitude of the component of the magnetic field perpendicular to the tether, which, in this case, is the vertical component of the Earth's magnetic field at this location:

$$B_{\text{vertical}} = B_\perp = \frac{\mathcal{E}}{\ell v} = \frac{1.17\ \text{V}}{(25.0\ \text{m})(7.80 \times 10^3\ \text{m/s})}$$

$$= 6.00 \times 10^{-6}\ \text{T} = \boxed{6.00\ \mu\text{T}}$$

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(b) Yes. The magnitude and direction of the Earth's field varies from one location to the other, so the induced voltage in the wire changes. Furthermore, the voltage will change if the tether cord or its velocity changes their orientations relative to the Earth's field.

(c) Either the long dimension of the tether or the velocity vector could be parallel to the magnetic field at some instant.

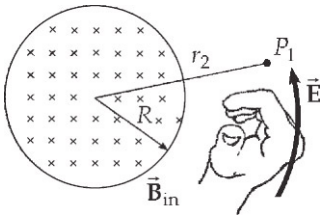
Section 31.4 Induced emf and Electric Fields

P31.39 Point P_1 lies outside the region of the uniform magnetic field. The rate of change of the field, in teslas per second, is

$$\frac{dB}{dt} = \frac{d}{dt}\left(2.00t^3 - 4.00t^2 + 0.800\right) = 6.00t^2 - 8.00t$$

where t is in seconds. At $t = 2.00$ s, we see that the field is increasing:

$$\frac{dB}{dt} = 6.00(2.00)^2 - 8.00(2.00) = 8.00 \text{ T/s}$$



ANS. FIG. P31.39

The magnetic flux is increasing into the page; therefore, by the right-hand rule (see figure), the induced electric field lines are counter-clockwise. [Also, if a conductor of radius r_1 were placed concentric with the field region, by Lenz's law, the induced current would be counterclockwise. Therefore, the direction of the induced electric field lines are counterclockwise.] The electric field at point P_1 is tangent to the electric field line passing through it.

(a) The magnitude of the electric field is (refer to Section 31.4 and Equation 31.8)

$$|E| = \frac{r}{2}\frac{dB}{dt} = \frac{r}{2}\left(6.00t^2 - 8.00t\right)$$

$$= \frac{0.050\,0}{2}\left[6.00(2.00)^2 - 8.00(2.00)\right] = 0.200 \text{ N/C}$$

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The magnitude of the force on the electron is

$$F = qE = eE = \left(1.60 \times 10^{-19}\ \text{C}\right)\left(0.200\ \text{N/C}\right) = \boxed{3.20 \times 10^{-20}\ \text{N}}$$

(b) Because the electron holds a negative charge, the direction of the force is opposite to the field direction. The force is tangent to the electric field line passing through at point P_1 and clockwise.

(c) The force is zero when the rate of change of the magnetic field is zero:

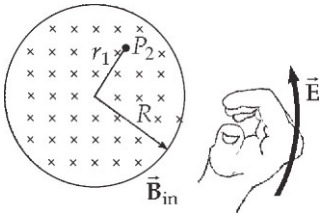
$$\frac{dB}{dt} = 6.00t^2 - 8.00t = 0 \rightarrow t = \boxed{0} \text{ or } t = \frac{8.00}{6.00} = \boxed{1.33\ \text{s}}$$

P31.40 Point P_2 lies inside the region of the uniform magnetic field. The rate of change of the field, in teslas per second, is

$$\frac{dB}{dt} = \frac{d}{dt}\left(0.030\ 0t^2 + 1.40\right) = 0.060\ 0t$$

where t is in seconds. At $t = 3.00$ s, we see that the field is increasing:

$$\frac{dB}{dt} = 0.060\ 0(3.00) = 0.180\ \text{T/s}$$



ANS. FIG. P31.40

The magnetic flux is increasing into the page; therefore, by the right-hand rule (see figure), the induced electric field lines are counterclockwise. The electric field at point P_2 is tangent to the electric field line passing through it.

(a) The situation is similar to that of Example 31.7.

$$\oint \vec{\mathbf{E}} \cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$$

$$E2\pi r = -\frac{d\Phi_B}{dt} = -\frac{d\left(B\pi R^2\right)}{dt} = -\pi R^2 \frac{dB}{dt}$$

$$E = -\frac{R^2}{2r}\left(0.060\ 0t\right)$$

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For $r = r_2 = 0.020\ 0$ m,

$$|E| = \frac{R^2}{2r}(0.060\ 0t)$$

$$= \frac{(0.025\ 0\ \text{m})^2}{2(0.020\ 0\ \text{m})}\left[0.060\ 0(3.00)\right] = \boxed{2.81\times10^{-3}\ \text{N/C}}$$

(b) The field is tangent to the electric field line passing through at point P_2 and counterclockwise.

P31.41 A problem similar to this is discussed in Example 31.7.

(a) $\oint \vec{\mathbf{E}}\cdot d\vec{\ell} = \left|\frac{d\Phi_B}{dt}\right|$ where $\Phi_B = BA = \mu_0 nI\left(\pi r^2\right)$

$$2\pi rE = \mu_0 n\left(\pi r^2\right)\frac{dI}{dt}$$

$$2\pi rE = \mu_0 n\left(\pi r^2\right)\frac{d}{dt}(5.00\sin 100\pi t)$$

$$= \mu_0 n\left(\pi r^2\right)(5.00)(100\pi)\cos 100\pi t$$

Solving for the electric field gives

$$E = \frac{\mu_0 n\left(\pi r^2\right)(5.00)(100\pi)(\cos 100\pi t)}{2\pi r}$$

$$= 250\mu_0 n\pi r\cos 100\pi t$$

Substituting numerical values and suppressing units,

$$E = 250\left(4\pi\times10^{-7}\right)\left(1.00\times10^{3}\right)\pi(0.0100)\cos 100\pi t$$

$$= \left(9.87\times10^{-3}\right)\cos 100\pi t$$

$\boxed{E = 9.87\cos 100\pi t \text{ where } E \text{ is in millivolts/meter and } t \text{ is in seconds.}}$

(b) If a viewer looks at the solenoid along its axis, and if the current is increasing in the counterclockwise direction, the magnetic flux is increasing toward the viewer; the electric field always opposes increasing magnetic flux; therefore, by the right-hand rule, the electric field lines are $\boxed{\text{clockwise}}$.

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Section 31.5 Generators and Motors

P31.42 (a) Use Equation 31.11, where B is the horizontal component of the magnetic field because the coil rotates about a vertical axis:

$$\begin{aligned}\mathcal{E}_{\max} &= NB_{\text{horizontal}}A\omega \\ &= 100\left(2.00\times10^{-5}\text{ T}\right)(0.200\text{ m})^2 \\ &\qquad \times\left[\left(1500\ \frac{\text{rev}}{\text{min}}\right)\left(\frac{2\pi\text{ rad}}{1\text{ rev}}\right)\left(\frac{1\text{ min}}{60\text{ s}}\right)\right] \\ &= 1.26\times10^{-2}\text{ V} = \boxed{12.6\text{ mV}}\end{aligned}$$

(b) Maximum emf occurs when the magnetic flux through the coil is changing the fastest. This occurs at the moment when the flux is zero, which is when the plane of the coil is parallel to the magnetic field.

P31.43 The emf induced in a rotating coil is directly proportional to the angular speed of the coil. Thus,

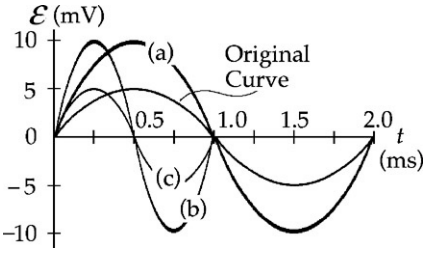
$$\frac{\mathcal{E}_2}{\mathcal{E}_1} = \frac{\omega_2}{\omega_1}$$

or $$\mathcal{E}_2 = \left(\frac{\omega_2}{\omega_1}\right)\mathcal{E}_1 = \left(\frac{500\text{ rev/min}}{900\text{ rev/min}}\right)(24.0\text{ V}) = \boxed{13.3\text{ V}}$$

P31.44 The induced emf is proportional to the number of turns and the angular speed.

(a) Doubling the number of turns has this effect:

amplitude doubles and period is unchanged



ANS FIG. P31.44

(b) Doubling the angular velocity has this effect:

doubles the amplitude and cuts the period in half

(c) Doubling the angular velocity while reducing the number of turns to one half the original value has this effect:

amplitude unchanged and period is cut in half

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P31.45 For the alternator,

$$\omega = (3\,000 \text{ rev/min})\left(\frac{2\pi \text{ rad}}{1 \text{ rev}}\right)\left(\frac{1 \text{ min}}{60 \text{ s}}\right) = 314 \text{ rad/s}$$

so

$$\mathcal{E} = -N\frac{d\Phi_B}{dt} = -250\frac{d}{dt}\left[\left(2.50\times10^{-4}\right)\cos(314t)\right]$$
$$= +250\left(2.50\times10^{-4}\right)(314)\sin(314t)$$

(a) $\mathcal{E} = 19.6\sin(314t)$ where $\mathcal{E}$ is in volts and t is in seconds.

(b) $\boxed{\mathcal{E}_{\max} = 19.6 \text{ V}}$

P31.46 Think of the semicircular conductor as enclosing half a coil of area $A = \frac{1}{2}\pi R^2$. There is no emf induced in the conductor until the magnetic flux through the area of the coil begins to change. The conductor is in the field region for only half a turn, so the flux changes over half a period $\frac{1}{2}T = \frac{1}{2}\left(\frac{2\pi}{\omega}\right) = \frac{\pi}{\omega}$. If we consider $t = 0$ to correspond to the time when the conductor is in the position shown in Figure P31.46 of the textbook, then there is no change in flux for a quarter of a turn, from $t =$ 0 to $t = \pi/2\omega$, then the flux has a periodic behavior

$\Phi_B = AB\cos\omega t = \frac{1}{2}\pi R^2 B\cos\omega t$ for a half a turn, from $t = \pi/2\omega$ to $t = 3\pi/2\omega$, then there is no change in flux for the final quarter of a turn, from $t = 3\pi/2\omega$ to $t = 2\pi/\omega$, at the end of which the coil has returned to its starting position. While in the field region, the induced emf is

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{1}{2}\pi R^2 B\frac{d}{dt}\cos\omega t = \frac{1}{2}\pi R^2\omega B\sin\omega t = \mathcal{E}_{\max}\sin\omega t$$

(a) The maximum emf is

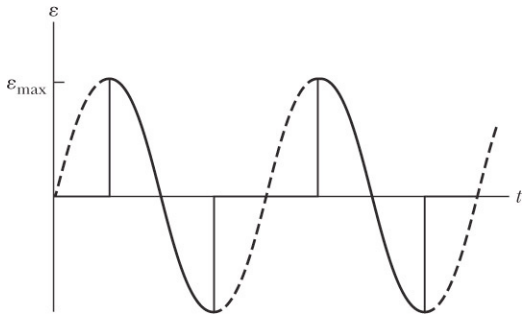
$$\mathcal{E}_{\max} = \frac{1}{2}\omega\pi R^2 B$$
$$= \frac{1}{2}\left[\left(\frac{120 \text{ rev}}{\min}\right)\left(\frac{2\pi \text{ rad}}{\text{rev}}\right)\left(\frac{1 \text{ min}}{60 \text{ s}}\right)\right]\pi(0.250 \text{ m})^2(1.30 \text{ T})$$
$$= \boxed{1.60 \text{ V}}$$

(b) During the time period that the coil travels in the field region, the emf varies as $\mathcal{E}_{\max}\sin\omega t$ for half a period, from $+\mathcal{E}_{\max}$, at $t = \pi/2\omega$, to $-\mathcal{E}_{\max}$, at $t = 3\pi/2\omega$; therefore, the average emf is $\boxed{\text{zero}}$.

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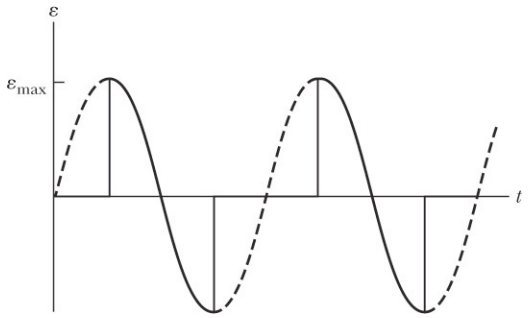
(c) The flux could also be written as $\Phi_B = \frac{1}{2}\pi R^2 B\cos\omega t$ so that it is a maximum at $t = 0$, but, in this case, the time period over which the flux changes would be from $t = 0$ to $t = 2\pi/\omega$, and the amplitude of the emf and its average would be the same as in the previous case; therefore, no change in either answer.

(d) The graph is



ANS. FIG. P31.46(d)

(e) If the time axis is chose so that the maximum emf occurs at the same time as it does in the figure of part (d) the graph is



ANS. FIG. P31.46(e)

P31.47 The magnetic field of the solenoid is given by

$$B = \mu_0 nI = \left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(200\ \text{m}^{-1}\right)(15.0\ \text{A})$$
$$= 3.77\times10^{-3}\ \text{T}$$

For the small coil, $\Phi_B = N\vec{\mathbf{B}}\cdot\vec{\mathbf{A}} = NBA\cos\omega t = NB\left(\pi r^2\right)\cos\omega t.$

Thus, $$\mathcal{E} = -\frac{d\Phi_B}{dt} = NB\pi r^2\omega\sin\omega t$$

Substituting numerical values,

$$\mathcal{E} = (30.0)\left(3.77\times10^{-3}\ \text{T}\right)\pi\left(0.080\,0\ \text{m}\right)^2\left(4.00\pi\ \text{s}^{-1}\right)\sin\left(4.00\pi t\right)$$
$$= \boxed{(28.6\ \text{mV})\sin\left(4.00\pi t\right)}$$

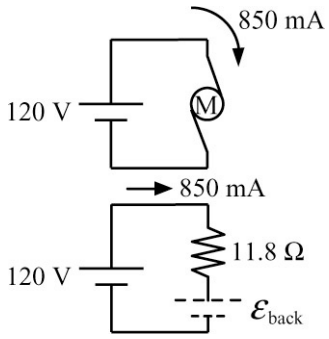
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P31.48 To analyze the actual circuit, we model it as the lower circuit diagram in ANS. FIG. P31.48.

(a) Kirchhoff's loop rule gives

$$+120\text{ V}-(0.850\text{ A})(11.8\ \Omega)-\mathcal{E}_{\text{back}}=0$$

$$\rightarrow \quad \mathcal{E}_{\text{back}}=\boxed{110\text{ V}}$$



ANS. FIG. P31.48

(b) The resistor is the device changing electrical work input into internal energy:

$$P=I^2R=(0.850\text{ A})^2(11.8\ \Omega)=\boxed{8.53\text{ W}}$$

(c) With no motion, the motor does not function as a generator, and $\mathcal{E}_{\text{back}}=0$. Then

$$120\text{ V}-I_c(11.8\ \Omega)=0\rightarrow I_c=10.2\text{ A}$$

$$P_c=I_c^2R=(10.2\text{ A})^2(11.8\ \Omega)=\boxed{1.22\text{ kW}}$$

P31.49 (a) The flux through the loop is

$$\Phi_B=BA\cos\theta=BA\cos\omega t$$

$$=(0.800\text{ T})\left(0.010\,0\text{ m}^2\right)\cos 2\pi(60.0)t$$

$$=\boxed{\left(8.00\text{ mT}\cdot\text{m}^2\right)\cos(377t)}$$

(b) $\mathcal{E}=-\dfrac{d\Phi_B}{dt}=\boxed{(3.02\text{ V})\sin(377t)}$

(c) $I=\dfrac{\mathcal{E}}{R}=\boxed{(3.02\text{ A})\sin(377t)}$

(d) $P=I^2R=\boxed{(9.10\text{ W})\sin^2(377t)}$

(e) $P=Fv=\tau\omega$ so $\tau=\dfrac{P}{\omega}=\boxed{(24.1\text{ mN}\cdot\text{m})\sin^2(377t)}$

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Section 31.6 Eddy Currents

P31.50 The current in the magnet creates an  upward magnetic field, so the N and S poles on the solenoid core are shown correctly. On the rail in front of the brake, the upward flux of $\vec{\mathbf{B}}$ increases as the coil approaches, so a current is induced here to create a downward magnetic field. This is  clockwise current, so the S pole on the rail is shown correctly. On the rail behind the brake, the upward magnetic flux is decreasing. The induced current in the rail will produce upward magnetic field by being  counterclockwise as the picture correctly shows.

Additional Problems

***P31.51** (a) From Faraday's law of induction,

$$|\mathcal{E}| = \frac{d\Phi_B}{dt} = \frac{d}{dt}(BA\cos\theta) = \frac{d}{dt}(BA) = A\frac{dB}{dt}$$

$$= \pi(0.060\ 0\ \text{m})^2\left(1.00\times10^4\ \text{T/s}\right)$$

$$= \boxed{113\ \text{V}}$$

(b) From Section 31.4, the electric field induced along the circumference of the circular area is given by

$$E = \frac{\mathcal{E}}{2\pi r} = \frac{113\ \text{V}}{2\pi(0.060\ 0\ \text{m})} = \boxed{300\ \text{V/m}}$$

***P31.52** Suppose we wrap twenty turns of wire into a flat compact circular coil of diameter 3 cm. Suppose we use a bar magnet to produce field $10^{-3}\ \text{T}$ through the coil in one direction along its axis. Suppose we then flip the magnet to reverse the flux in 10^{-1} s. The average induced emf is then

$$\bar{\mathcal{E}} = -N\frac{\Delta\Phi_B}{\Delta t} = -N\frac{\Delta[BA\cos\theta]}{\Delta t} = -NB(\pi r^2)\left(\frac{\cos180^\circ - \cos0^\circ}{\Delta t}\right)$$

$$\bar{\mathcal{E}} = -(20)(10^{-3}\ \text{T})\pi(0.015\ 0\ \text{m})^2\left(\frac{-2}{10^{-1}\ \text{s}}\right)\boxed{\sim10^{-4}\ \text{V}}$$

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***P31.53** The magnitude of the average emf is given by

$$|\bar{\mathcal{E}}| = N\frac{|\Delta\Phi_B|}{\Delta t} = \frac{NBA(\Delta\cos\theta)}{\Delta t}$$

$$= \frac{200(1.1\text{ T})\left(100\times10^{-4}\text{ m}^2\right)|\cos 180° - \cos 0°|}{0.10\text{ s}} = 44\text{ V}$$

The average current induced in the coil is therefore

$$I = \frac{|\mathcal{E}|}{R} = \frac{44\text{ V}}{5.0\ \Omega} = \boxed{8.8\text{ A}}$$

P31.54 (a) If the magnetic field were increasing, the flux would be increasing out of the page, so the induced current would tend to oppose the increase by generating a field into the page. The direction of such a current would be clockwise. This is the case here, so the field is $\boxed{\text{increasing}}$.

(b) The normal to the enclosed area can be taken to be parallel to the magnetic field, so the flux through the loop is

$$\Phi_B = BA\cos 0.00° = BA$$

The rate of change of the flux is

$$\frac{d\Phi_B}{dt} = \frac{d}{dt}(BA\cos 0.00°) = A\frac{dB}{dt}$$

and the induced emf is

$$|\mathcal{E}| = \left|-\frac{d\Phi_B}{dt}\right| \quad \rightarrow \quad IR = A\frac{dB}{dt} = \pi r^2\frac{dB}{dt}$$

Therefore,

$$\frac{dB}{dt} = \frac{IR}{\pi r^2} = \frac{\left(2.50\times10^{-3}\text{ A}\right)(0.500\ \Omega)}{\pi(0.080\ 0\text{ m})^2}$$

$$= 0.062\ 2\text{ T/s}$$

$$= \boxed{62.2\text{ mT/s}}$$

P31.55 The emf through the hoop is given by

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -A\frac{dB}{dt} = -0.160\frac{d}{dt}\left(0.350e^{-t/200}\right)$$

$$= \frac{(1.60)(0.350)}{200}e^{-t/200}$$

where $\mathcal{E}$ is in volts and t in seconds. For t = 4.00 s,

$$\mathcal{E} = \frac{\left(0.160\text{ m}^2\right)(0.350\text{ T})}{2.00\text{ s}}e^{-4.00/2.00} = \boxed{3.79\text{ mV}}$$

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P31.56 The emf through the hoop is given by

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -A\frac{dB}{dt} = -A\frac{d}{dt}\left(B_{\max}e^{-t/\tau}\right) = \boxed{\frac{AB_{\max}}{\tau}e^{-t/\tau}}$$

P31.57 $$\mathcal{E} = -N\frac{\Delta}{\Delta t}(BA\cos\theta) = -N\left(\pi r^2\right)\cos 0^\circ\frac{\Delta B}{\Delta t}$$

$$= -1\left(0.005\ 00\ \text{m}^2\right)(1)\left(\frac{1.50\ \text{T} - 5.00\ \text{T}}{20.0\times10^{-3}\ \text{s}}\right) = 0.875\ \text{V}$$

(a) $$I = \frac{\mathcal{E}}{R} = \frac{0.875\ \text{V}}{0.020\ 0\ \Omega} = \boxed{43.8\ \text{A}}$$

(b) $$P = \mathcal{E}I = (0.875\ \text{V})(43.8\ \text{A}) = \boxed{38.3\ \text{W}}$$

P31.58 (a) Motional emf produces a current $I = \frac{\mathcal{E}}{R} = \boxed{\frac{B\ell v}{R}}$.

(b) $\boxed{\text{Particle in equilibrium}}$

(c) The circuit encloses increasing flux of magnetic field into the page, so it tries to make its own field out of the page, by carrying counterclockwise current. The current flows upward in the bar, so the magnetic field produces a backward magnetic force $F_B = I\ell B$ (to the left) on the bar. This force increases until the bar has reached a speed when the backward force balances the applied force F:

$$F = F_B = I\ell B = \frac{\mathcal{E}}{R}\ell B = \frac{(B\ell v)}{R}\ell B = \frac{B^2\ell^2}{R}v$$

$$v = \frac{FR}{B^2\ell^2} = \frac{(0.600\ \text{N})(48.0\ \Omega)}{(0.400\ \text{T})^2(0.800\ \text{m})^2} = \boxed{281\ \text{m/s}}$$

(d) $$I = \frac{\mathcal{E}}{R} = \frac{B\ell v}{R} = \frac{B\ell}{R}\frac{FR}{B^2\ell^2} = \frac{F}{B\ell} = \frac{0.600\ \text{N}}{(0.400\ \text{T})(0.800\ \text{m})} = \boxed{1.88\ \text{A}}$$

(e) $$P = I^2R = \left(\frac{F}{B\ell}\right)^2 R = \left[\frac{0.600\ \text{N}}{(0.400\ \text{T})(0.800\ \text{m})}\right]^2(48.0\ \Omega) = \boxed{169\ \text{W}}$$

(f) $$P = Fv = F\frac{FR}{B^2\ell^2} = \frac{F^2R}{B^2\ell^2} = \frac{(0.600\ \text{N})^2(48.0\ \Omega)}{(0.400\ \text{T})^2(0.800\ \text{m})^2} = \boxed{169\ \text{W}}$$

(g) $\boxed{\text{Yes.}}$

(h) $\boxed{\text{Increase}}$ because the speed is proportional to the resistance, as shown in part (c).

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(i) Yes.

(j) Larger because the speed is greater.

P31.59 $$\mathcal{E} = -N\frac{d}{dt}(BA\cos\theta) = -N\left(\pi r^2\right)\cos 0°\left(\frac{dB}{dt}\right)$$

$$\mathcal{E} = -(30.0)\left[\pi\left(2.70\times10^{-3}\right)^2\right](1)$$

$$\times\frac{d}{dt}\left[50.0\times10^{-3} + \left(3.20\times10^{-3}\right)\sin(1\ 046\pi t)\right]$$

$$\mathcal{E} = -(30.0)\left[\pi\left(2.70\times10^{-3}\right)^2\right]\left[\left(3.20\times10^{-3}\right)(1\ 046\pi)\cos(1\ 046\pi t)\right]$$

$$= -\left(7.22\times10^{-3}\right)\cos(1\ 046\pi t)$$

$\mathcal{E} = -7.22\cos(1\ 046\pi t)$ where $\mathcal{E}$ is in millivolts and t is in seconds.

P31.60 Model the loop as a particle under a net force. The two forces on the loop are the gravitational force in the downward direction and the magnetic force in the upward direction. The magnetic force arises from the current generated in the loop due to the motion of its lower edge through the magnetic field. As the loop falls, the motional emf $\mathcal{E} = Bwv$ induced in the bottom side of the loop produces a current $I = Bwv/R$ in the loop. From Newton's second law,

$$\sum F_y = ma_y \rightarrow F_B - F_g = Ma_y \rightarrow IwB - Mg = Ma_y$$

$$\rightarrow \left(\frac{Bwv}{R}\right)wB - Mg = Ma_y \rightarrow \frac{B^2w^2v}{MR} - g = a_y$$

The largest possible value of v, the terminal speed v_T, will occur when $a_y = 0$. Set $a_y = 0$ and solve for the terminal speed:

$$\frac{B^2w^2v_T}{MR} - g = 0 \quad \rightarrow \quad v_T = \frac{MgR}{B^2w^2}$$

Substituting numerical values,

$$v_T = \frac{(0.100\ \text{kg})(9.80\ \text{m/s}^2)(1.00\ \Omega)}{(1.00\ \text{T})^2(0.500\ \text{m})^2} = 3.92\ \text{m/s}$$

This is the highest speed the loop can have while the upper edge is above the field, so it cannot possibly be moving at 4.00 m/s.

P31.61 For a counterclockwise trip around the left-hand loop, with $B = At$,

$$\frac{d}{dt}\left[At\left(2a^2\right)\cos 0°\right] - I_1(5R) - I_{PQ}R = 0$$

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and for the right-hand loop,

$$\frac{d}{dt}\left[Ata^2\right] + I_{PQ}R - I_2(3R) = 0$$

where $I_{PQ} = I_1 - I_2$ is the upward current in QP.

Thus, $2Aa^2 - 5R\left(I_{PQ} + I_2\right) - I_{PQ}R = 0$

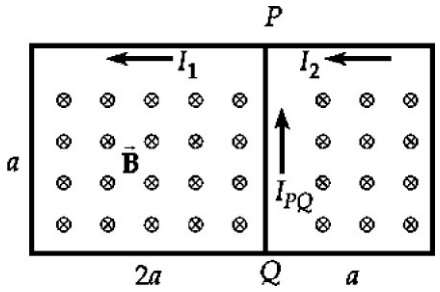
and $Aa^2 + I_{PQ}R = I_2(3R)$

$$2Aa^2 - 6RI_{PQ} - \frac{5}{3}\left(Aa^2 + I_{PQ}R\right) = 0$$

solving, $I_{PQ} = \dfrac{Aa^2}{23R}$ upward

and since $R = (0.100\ \Omega/\text{m})(0.650\ \text{m}) = 0.650\ 0\ \Omega$,

$$I_{PQ} = \frac{\left(1.00\times10^{-3}\ \text{T/s}\right)(0.650\ \text{m})^2}{23(0.065\ 0\ \Omega)} = \boxed{283\ \mu\text{A upward}}$$



ANS. FIG. P31.61

P31.62 (a) $I = \dfrac{dq}{dt} = \dfrac{\mathcal{E}}{R}$ where $\mathcal{E} = -N\dfrac{d\Phi_B}{dt}$ so $\displaystyle\int dq = \frac{N}{R}\int_{\Phi_1}^{\Phi_2} d\Phi_B$

and the charge passing any point in the circuit will be $|Q| = \dfrac{N}{R}(\Phi_2 - \Phi_1)$.

(b) $Q = \dfrac{N}{R}\left[BA\cos 0 - BA\cos\left(\dfrac{\pi}{2}\right)\right] = \dfrac{BAN}{R}$

so $B = \dfrac{RQ}{NA} = \dfrac{(200\ \Omega)\left(5.00\times10^{-4}\ \text{C}\right)}{(100)\left(40.0\times10^{-4}\ \text{m}^2\right)} = \boxed{0.250\ \text{T}}$

P31.63 The emf induced between the ends of the moving bar is

$$\mathcal{E} = B\ell v = (2.50\ \text{T})(0.350\ \text{m})\left(8.00\ \text{m/s}\right) = 7.00\ \text{V}$$

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The left-hand loop contains decreasing flux away from you, so the induced current in it will be  clockwise, to produce its own field directed away from you. Let I_1 represent the current flowing upward through the 2.00-Ω resistor. The right-hand loop will carry counterclockwise current. Let I_3 be the upward current in the 5.00-Ω resistor.

(a) Kirchhoff's loop rule then gives:

$$+7.00\text{ V} - I_1(2.00\ \Omega) = 0 \qquad \text{or} \qquad I_1 = \boxed{3.50\text{ A}}$$

$$\text{and} \quad +7.00\text{ V} - I_3(5.00\ \Omega) = 0 \qquad \text{or} \qquad I_3 = \boxed{1.40\text{ A}}$$

(b) The total power converted in the resistors of the circuit is

$$P = \mathcal{E}I_1 + \mathcal{E}I_3 = \mathcal{E}\left(I_1 + I_3\right) = (7.00\text{ V})(3.50\text{ A} + 1.40\text{ A})$$

$$= \boxed{34.3\text{ W}}$$

(c) *Method 1*: The current in the sliding conductor is downward with value $I_2 = 3.50\text{ A} + 1.40\text{ A} = 4.90\text{ A}$. The magnetic field exerts a force of $F_m = I\ell B = (4.90\text{ A})(0.350\text{ m})(2.50\text{ T}) = 4.29\text{ N}$ directed  toward the right on this conductor. An outside agent must then exert a force of $\boxed{4.29\text{ N}}$ to the left to keep the bar moving.

Method 2: The agent moving the bar must supply the power according to $P = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}} = Fv\cos 0°$. The force required is then:

$$F = \frac{P}{v} = \frac{34.3\text{ W}}{8.00\text{ m/s}} = \boxed{4.29\text{ N}}$$

P31.64 The enclosed flux is $\Phi_B = BA = B\pi r^2$. The particle moves according to

$$\sum \vec{\mathbf{F}} = m\vec{\mathbf{a}}: \quad qvB\sin 90° = \frac{mv^2}{r} \quad \rightarrow \quad r = \frac{mv}{qB}$$

Thus, $$\Phi_B = \frac{B\pi m^2 v^2}{q^2 B^2}.$$

(a) $$v = \sqrt{\frac{\Phi_B q^2 B}{\pi m^2}} = \sqrt{\frac{\left(15\times 10^{-6}\text{ T}\cdot\text{m}^2\right)\left(30\times 10^{-9}\text{ C}\right)^2 (0.6\text{ T})}{\pi\left(2\times 10^{-16}\text{ kg}\right)^2}}$$

$$= \boxed{2.54\times 10^5\text{ m/s}}$$

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(b) Energy for the particle-electric field system is conserved in the firing process:

$$U_i = K_f: \quad q\Delta V = \frac{1}{2}mv^2$$

From which we obtain

$$\Delta V = \frac{mv^2}{2q} = \frac{\left(2\times10^{-16}\ \text{kg}\right)\left(2.54\times10^{5}\ \text{m/s}\right)^2}{2\left(30\times10^{-9}\ \text{C}\right)} = \boxed{215\ \text{V}}$$

P31.65 The normal to the loop is horizontally north, at 35.0° to the magnetic field. We assume that 0.500 Ω is the total resistance around the circuit, including the ammeter.

$$Q = \int I dt = \int \frac{\mathcal{E} dt}{R} = \frac{1}{R}\int -\left(\frac{d\Phi_B}{dt}\right)dt = -\frac{1}{R}\int d\Phi_B$$

$$= -\frac{1}{R}\int d(BA\cos\theta) = -\frac{B\cos\theta}{R}\int_{A_1=a^2}^{A_2=0} dA$$

$$Q = -\left[\frac{B\cos\theta}{R}A\right]_{A_1=a^2}^{A_2=0} = \frac{B\cos\theta\, a^2}{R}$$

$$= \frac{(35.0\times10^{-6}\ \text{T})(\cos 35.0°)(0.200\ \text{m})^2}{0.500\ \Omega}$$

$$= \boxed{2.29\times10^{-6}\ \text{C}}$$

P31.66 (a) To find the induced current, we first compute the induced emf,

$$\mathcal{E} = B\ell v = (0.0800\ \text{T})(1.50\ \text{m})(3.00\ \text{m/s}) = 0.360\ \text{V}.$$

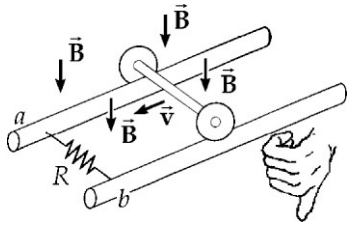
Then,

$$I = \frac{\mathcal{E}}{R} = \frac{0.360\ \text{V}}{0.400\ \Omega} = \boxed{0.900\ \text{A}}$$

(b) The applied force must balance the magnetic force

$$F = F_B = I\ell B$$

$$= (0.900\ \text{A})(1.50\ \text{m})(0.0800\ \text{T}) = \boxed{0.108\ \text{N}}$$



ANS. FIG. P31.66

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(c) Since the magnetic flux $\vec{\mathbf{B}} \cdot \vec{\mathbf{A}}$ between the axle and the resistor is in effect decreasing, the induced current is clockwise so that it produces a downward magnetic field to oppose the decrease in flux: thus, current flows through R from b to a. $\boxed{\text{Point } b}$ is at the higher potential.

(d) $\boxed{\text{No}}$. Magnetic flux will increase through a loop between the axle and the resistor to the left of ab. Here counterclockwise current will flow to produce an upward magnetic field to oppose the increase in flux. The current in R is still from b to a.

***P31.67** (a) From Equation 31.3, the emf induced in the loop is given by

$$\mathcal{E} = -N\frac{d}{dt}BA\cos\theta = -1\frac{d}{dt}\left(B\frac{\theta a^2}{2}\cos 0^\circ\right)$$

$$= -\frac{Ba^2}{2}\frac{d\theta}{dt} = -\frac{1}{2}Ba^2\omega$$

Substituting numerical values,

$$\mathcal{E} = -\frac{1}{2}(0.500\text{ T})(0.500\text{ m})^2(2.00\text{ rad/s})$$

$$= -0.125\text{ V} = \boxed{0.125\text{ V clockwise}}$$

The minus sign indicates that the induced emf produces clockwise current, to make its own magnetic field into the page.

(b) At this instant,

$$\theta = \omega t = (2.00\text{ rad/s})(0.250\text{ s}) = 0.500\text{ rad}$$

The arc PQ has length

$$r\theta = (0.500\text{ rad})(0.500\text{ m}) = 0.250\text{ m}$$

The length of the circuit is

$$0.500\text{ m} + 0.500\text{ m} + 0.250\text{ m} = 1.25\text{ m}$$

Its resistance is

$$(1.25\text{ m})(5.00\ \Omega/\text{m}) = 6.25\ \Omega$$

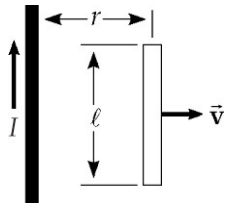
The current is then

$$I = \frac{\mathcal{E}}{R} = \frac{0.125\text{ V}}{6.25\ \Omega} = \boxed{0.020\,0\text{ A clockwise}}$$

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P31.68 At a distance r from wire, $B = \dfrac{\mu_0 I}{2\pi r}$. Using $\mathcal{E} = B\ell v$, we find that

$$|\mathcal{E}| = \frac{\mu_0 v I \ell}{2\pi r}$$



ANS. FIG. P31.68

P31.69 (a) We use $\mathcal{E} = -N\dfrac{\Delta\Phi_B}{\Delta t}$, with $N = 1$.

Taking $a = 5.00 \times 10^{-3}$ m to be the radius of the washer, and $h = 0.500$ m, the change in flux through the washer from the time it is released until it hits the tabletop is

$$\Delta\Phi_B = B_f A - B_i A = A\left(B_f - B_i\right) = \pi a^2\left(\frac{\mu_0 I}{2\pi(h+a)} - \frac{\mu_0 I}{2\pi a}\right)$$

$$= \frac{a^2\mu_0 I}{2}\left(\frac{1}{h+a} - \frac{1}{a}\right) = \frac{-\mu_0 a h I}{2(h+a)}$$

The time for the washer to drop a distance h (from rest) is: $\Delta t = \sqrt{\dfrac{2h}{g}}$. Therefore,

$$\mathcal{E} = -\frac{\Delta\Phi_B}{\Delta t} = \frac{\mu_0 a h I}{2(h+a)\Delta t} = \frac{\mu_0 a h I}{2(h+a)}\sqrt{\frac{g}{2h}} = \frac{\mu_0 a I}{2(h+a)}\sqrt{\frac{gh}{2}}$$

Substituting numerical values,

$$\mathcal{E} = \frac{\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)\left(5.00\times10^{-3}\ \text{m}\right)(10.0\ \text{A})}{2(0.500\ \text{m} + 0.005\,00\ \text{m})} \times\sqrt{\frac{(9.80\ \text{m/s}^2)(0.500\ \text{m})}{2}}$$

$$= \boxed{97.4\ \text{nV}}$$

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(b) Since the magnetic flux going through the washer (into the plane of the page in the figure) is decreasing in time, a current will form in the washer so as to oppose that decrease. To oppose the decrease, the magnetic field from the induced current also must point into the plane of the page. Therefore, the current will flow in a clockwise direction.

P31.70 (a) We would need to know whether the field is increasing or decreasing.

(b) To find the resistance at maximum power, we note that

$$P = \mathcal{E}I = \frac{\mathcal{E}^2}{R} = \frac{\left(N\frac{dB}{dt}\pi r^2 \cos 0°\right)^2}{R}$$

Solving for the resistance then gives

$$R = \frac{\left(N\frac{dB}{dt}\pi r^2\right)^2}{P} = \frac{\left[220(0.020\ \text{T/s})\pi(0.120\ \text{m})^2\right]^2}{160\ \text{W}} = \boxed{248\ \mu\Omega}$$

(c) Higher resistance would reduce the power delivered.

P31.71 Let θ represent the angle between the perpendicular to the coil and the magnetic field. Then $\theta = 0$ at $t = 0$ and $\theta = \omega t$ at all later times.

(a) The emf induced in the coil is given by

$$\mathcal{E} = -N\frac{d}{dt}(BA\cos\theta) = -NBA\frac{d}{dt}(\cos\omega t) = +NBA\,\omega\sin\omega t$$

The maximum value of $\sin\theta$ is 1, so the maximum voltage is

$$\mathcal{E}_{\max} = NBA\omega = (60)(1.00\ \text{T})\left(0.020\ 0\ \text{m}^2\right)(30.0\ \text{rad/s})$$
$$= \boxed{36.0\ \text{V}}$$

(b) The rate of change of magnetic flux is

$$\frac{d\Phi_B}{dt} = \frac{d}{dt}(BA\cos\theta) = -BA\omega\sin\omega t$$

The minimum value of $\sin\theta$ is –1, so the maximum of $d\Phi_B/dt$ is

$$\left(\frac{d\Phi_B}{dt}\right)_{\max} = +BA\omega = (1.00\ \text{T})(0.020\ 0\ \text{m}^2)(30.0\ \text{rad/s})$$
$$= \boxed{0.600\ \text{T}\cdot\text{m}^2/\text{s}}$$

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(c) At $t = 0.050\ 0$ s,

$$\mathcal{E} = NBA\omega \sin \omega t = (36.0\text{ V})\sin\left[(30.0\text{ rad/s})(0.050\ 0\text{ s})\right]$$

$$= (36.0\text{ V})\sin(1.50\text{ rad}) = (36.0\text{ V})(\sin\ 85.9^\circ) = \boxed{35.9\text{ V}}$$

(d) The emf is maximum when $\theta = 90°$, and $\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{\mu}} \times \vec{\mathbf{B}}$, so

$$\tau_{\max} = \mu B \sin 90^\circ = NIAB = N\mathcal{E}_{\max}\frac{AB}{R}$$

and $$\tau_{\max} = (60)(36.0\text{ V})\frac{(0.020\ 0\text{ m}^2)(1.00\text{ T})}{10.0\ \Omega} = \boxed{4.32\text{ N}\cdot\text{m}}$$

P31.72 The emf induced in the loop is

$$\mathcal{E} = -\frac{d}{dt}(NBA) = -1\left(\frac{dB}{dt}\right)\pi a^2 = \pi a^2 K$$

(a) The charge on the fully-charged capacitor is

$$Q = C\mathcal{E} = \boxed{C\pi a^2 K}$$

(b) $\vec{\mathbf{B}}$ into the paper is decreasing; therefore, current will attempt to counteract this by producing a magnetic field into the page to oppose the decrease in flux. To do this, the current must be clockwise, so positive charge will go to the upper plate.

(c) The changing magnetic field through the enclosed area of the loop induces a clockwise electric field within the loop, and this causes electric force to push on charges in the wire.

P31.73 (a) The time interval required for the coil to move distance ℓ and exit the field is $\Delta t = \ell/v$, where v is the constant speed of the coil. Since the speed of the coil is constant, the flux through the area enclosed by the coil decreases at a constant rate. Thus, the instantaneous induced emf is the same as the average emf over the interval Δt, or

$$\mathcal{E} = -N\frac{\Delta\Phi}{\Delta t} = -N\frac{(0 - BA)}{t - 0} = N\frac{B\ell^2}{t} = \frac{NB\ell^2}{\ell/v} = \boxed{NB\ell v}$$

(b) The current induced in the coil is

$$I = \frac{\mathcal{E}}{R} = \boxed{\frac{NB\ell v}{R}}$$

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(c) The power delivered to the coil is given by $P = I^2R$, or

$$P = \left(\frac{N^2B^2\ell^2v^2}{R^2}\right)R = \boxed{\frac{N^2B^2\ell^2v^2}{R}}$$

(d) The rate that the applied force does work must equal the power delivered to the coil, so $F_{app} \cdot v = P$ or

$$F_{app} = \frac{P}{v} = \frac{N^2B^2\ell^2v^2/R}{v} = \boxed{\frac{N^2B^2\ell^2v}{R}}$$

(e) As the coil is emerging from the field, the flux through the area it encloses is directed into the page and decreasing in magnitude. Thus, the *change* in the flux through the coil is directed out of the page. The induced current must then flow around the coil in such a direction as to produce flux into the page through the enclosed area, opposing the change that is occurring. This means that the current must flow clockwise around the coil.

(f) As the coil is emerging from the field, the left side of the coil is carrying an induced current directed toward the top of the page through a magnetic field that is directed into the page. By the right-hand rule, this side of the coil will experience a magnetic force directed to the left, opposing the motion of the coil.

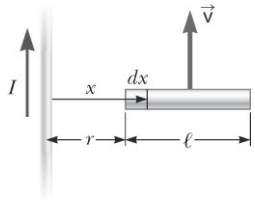
P31.74 The magnetic field at a distance x from wire is

$$B = \frac{\mu_0 I}{2\pi x}$$

The emf induced in an element in the bar of length dx is $|d\mathcal{E}| = Bvdx$. The total emf induced along the entire length of the bar is then

$$|\mathcal{E}| = \int_r^{r+\ell} Bv\,dx = \int_r^{r+\ell} \frac{\mu_0 I}{2\pi x} v\,dx = \frac{\mu_0 Iv}{2\pi}\int_r^{r+\ell}\frac{dx}{x} = \frac{\mu_0 Iv}{2\pi}\ln x\Big|_r^{r+\ell}$$

$$|\mathcal{E}| = \frac{\mu_0 Iv}{2\pi}\ln\left(\frac{r+\ell}{r}\right)$$



ANS. FIG. P31.74

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P31.75 We are given

$$\Phi_B = \left(6.00t^3 - 18.0t^2\right)$$

Thus,

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -18.0t^2 + 36.0t$$

Maximum $\mathcal{E}$ occurs when $\frac{d\mathcal{E}}{dt} = -36.0t + 36.0 = 0,$ which gives $t = 1.00$ s.

Therefore, the maximum current (at $t = 1.00$ s) is

$$I = \frac{\mathcal{E}}{R} = \frac{(-18.0 + 36.0)\ \text{V}}{3.00\ \Omega} = \boxed{6.00\ \text{A}}$$

P31.76 The magnetic field at a distance x from a long wire is $B = \frac{\mu_0 I}{2\pi x}$. We find an expression for the flux through the loop.

$$d\Phi_B = \frac{\mu_0 I}{2\pi x}(\ell dx)$$

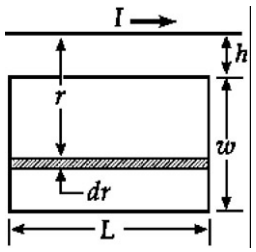
so $$\Phi_B = \frac{\mu_0 I\ell}{2\pi}\int_r^{r+w}\frac{dx}{x} = \frac{\mu_0 I\ell}{2\pi}\ln\left(1 + \frac{w}{r}\right)$$

Therefore,

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = \frac{\mu_0 I\ell v}{2\pi r}\frac{w}{(r+w)}$$

and $$I = \frac{\mathcal{E}}{R} = \boxed{\frac{\mu_0 I\ell v}{2\pi Rr}\frac{w}{(r+w)}}$$

P31.77 The magnetic field produced by the current in the straight wire is perpendicular to the plane of the coil at all points within the coil. At a distance r from the wire, the magnitude of the field is $B = \frac{\mu_0 I}{2\pi r}$. Thus, the flux through an element of length L and width dr is



ANS. FIG. P31.77

$$d\Phi_B = BLdr = \frac{\mu_0 IL}{2\pi}\frac{dr}{r}$$

The total flux through the coil is

$$\Phi_B = \frac{\mu_0 IL}{2\pi}\int_h^{h+w}\frac{dr}{r} = \frac{\mu_0 I_{\max} L}{2\pi}\ln\left(\frac{h+w}{h}\right)\sin\left(\omega t + \phi\right)$$

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Finally, the induced emf is

$$\mathcal{E} = -N\frac{d\Phi_B}{dt}$$

$$= -\frac{\mu_0 N I_{\max} L\omega}{2\pi}\ln\left(1+\frac{w}{h}\right)\cos(\omega t+\phi)$$

$$\mathcal{E} = -\frac{\left(4\pi\times10^{-7}\ \text{T}\cdot\text{m/A}\right)(100)(50.0\ \text{A})(0.200\ \text{m})(200\pi\ \text{rad/s})}{2\pi}$$

$$\times\ln\left(1+\frac{0.050\ 0\ \text{m}}{0.050\ 0\ \text{m}}\right)\cos(\omega t+\phi)$$

$$\mathcal{E} = \boxed{-87.1\cos(200\pi t+\phi)}\text{, where } \mathcal{E} \text{ is in millivolts and}$$

t is in seconds

The term $\sin(\omega t+\phi)$ in the expression for the current in the straight wire does not change appreciably when ωt changes by 0.10 rad or less. Thus, the current does not change appreciably during a time interval

$$\Delta t < \frac{0.10}{\left(200\pi\ \text{s}^{-1}\right)} = 1.6\times10^{-4}\ \text{s}$$

We define a critical length,

$$c\Delta t = \left(3.00\times10^{8}\ \text{m/s}\right)\left(1.6\times10^{-4}\ \text{s}\right) = 4.8\times10^{4}\ \text{m}$$

equal to the distance to which field changes could be propagated during an interval of 1.6×10^{-4} s. This length is so much larger than any dimension of the coil or its distance from the wire that, although we consider the straight wire to be infinitely long, we can also safely ignore the field propagation effects in the vicinity of the coil. Moreover, the phase angle can be considered to be constant along the wire in the vicinity of the coil.

If the angular frequency ω were much larger, say, $200\pi \times 10^5\ \text{s}^{-1}$, the corresponding critical length would be only 48 cm. In this situation propagation effects would be important and the above expression for $\mathcal{E}$ would require modification. As a general rule we can consider field propagation effects for circuits of laboratory size to be negligible for frequencies, $f = \dfrac{\omega}{2\pi}$, that are less than about 10^6 Hz.

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P31.78 (a) The induced emf is $\mathcal{E} = B\ell v$, where $B = \dfrac{\mu_0 I}{2\pi y}$, $\ell = 0.800$,

$v_f = v_i + gt = 9.80t$, and $y = y_f = y_i - \dfrac{1}{2}gt^2 = 0.800 - (4.90)t^2$ where I is in amperes, ℓ and y are in meters, v is in meters per second, and t in seconds.

Thus,

$$\mathcal{E} = \frac{\left(4\pi \times 10^{-7}\right)(200)}{2\pi\left(0.800 - 4.90t^2\right)}(0.300)(9.80)t = \boxed{\frac{\left(1.18 \times 10^{-4}\right)t}{0.800 - 4.90t^2}}$$

where $\mathcal{E}$ is in volts and t in seconds.

(b) The emf is zero when $t = 0$.

(c) As $0.800 - 4.90t^2 \rightarrow 0$, $t \rightarrow 0.404$ s and the emf diverges to infinity.

(d) At $t = 0.300$ s,

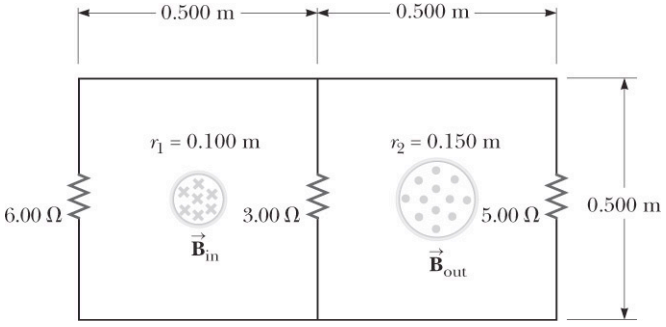
$$\mathcal{E} = \frac{\left(1.18 \times 10^{-4}\right)(0.300)}{\left[0.800 - 4.90(0.300)^2\right]}\ \text{V} = \boxed{98.3\ \mu\text{V}}$$

Challenge Problems

P31.79 In the loop on the left, the induced emf is

$$|\mathcal{E}| = \frac{d\Phi_B}{dt} = A\frac{dB}{dt} = \pi(0.100\ \text{m})^2(100\ \text{T/s}) = \pi\ \text{V}$$

and it attempts to produce a counterclockwise current in this loop.



ANS. FIG. P31.79

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In the loop on the right, the induced emf is

$$|\mathcal{E}| = \frac{d\Phi_B}{dt} = \pi(0.150\text{ m})^2(100\text{ T/s}) = 2.25\pi\text{ V}$$

and it attempts to produce a clockwise current. Assume that I_1 flows down through the 6.00-Ω resistor, I_2 flows down through the 5.00-Ω resistor, and that I_3 flows up through the 3.00-Ω resistor.

From Kirchhoff's junction rule: $I_3 = I_1 + I_2$ **[1]**

Using the loop rule on the left loop: $6.00I_1 + 3.00I_3 = \pi$ **[2]**

Using the loop rule on the right loop: $5.00I_2 + 3.00I_3 = 2.25\pi$ **[3]**

Solving these three equations simultaneously,

$$I_3 = \frac{(\pi - 3I_3)}{6} + \frac{(2.25\pi - 3I_3)}{5}$$

which then gives

$$I_1 = \boxed{0.062\ 3\text{ A}},\ I_2 = \boxed{0.860\text{ A}},\text{ and } I_3 = \boxed{0.923\text{ A}}$$

P31.80 (a) Consider an annulus of radius r, width dr, thickness b, and resistivity ρ. Around its circumference, a voltage is induced according to

$$\mathcal{E} = -N\frac{d}{dt}\vec{\mathbf{B}}\cdot\vec{\mathbf{A}} = -(1)\left[\frac{d}{dt}B_{\max}(\cos\omega t)\right]\pi r^2 = +B_{\max}\pi r^2\omega\sin\omega t$$

The resistance around the loop is $\frac{\rho\ell}{dA} = \frac{\rho(2\pi r)}{b\,dr}$. The eddy current in the ring is

$$dI = \frac{\mathcal{E}}{\text{resistance}} = \frac{B_{\max}\pi r^2\omega(\sin\omega t)}{\rho(2\pi r)/b\,dr} = \frac{B_{\max}rb\omega\sin\omega t}{2\rho}dr$$

The instantaneous power is

$$dP = \mathcal{E}\,dI = \frac{B_{\max}^2\pi r^3 b\omega^2\sin^2\omega t}{2\rho}dr$$

The time average of the function $\sin^2\omega t = \frac{1}{2} - \frac{1}{2}\cos 2\omega t$ is $\frac{1}{2} - 0 = \frac{1}{2}$, so the time-averaged power delivered to the annulus is

$$d\bar{P} = \frac{B_{\max}^2\pi r^3 b\omega^2}{4\rho}dr$$

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The average power delivered to the disk is

$$P = \int dP = \int_0^R \frac{B_{\max}^2 \pi b \omega^2}{4\rho} r^3 dr$$

$$P = \frac{B_{\max}^2 \pi b \omega^2}{4\rho}\left(\frac{R^4}{4} - 0\right) = \boxed{\frac{\pi B_{\max}^2 R^4 b \omega^2}{16\rho}}$$

(b) When $B_{\max}$ doubles, $B_{\max}^2$ and P become $\boxed{4}$ times larger.

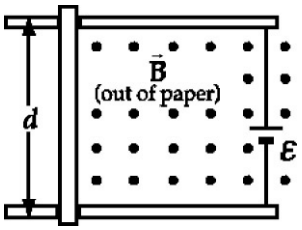
(c) When f doubles, $\omega = 2\pi f$ doubles, and ω^2 and P become $\boxed{4}$ times larger.

(d) When R doubles, R^4 and P become $2^4 = \boxed{16}$ times larger.

P31.81 The current in the rod is

$$I = \frac{\mathcal{E} + \mathcal{E}_{\text{induced}}}{R}$$

where $\mathcal{E}_{\text{induced}} = -Bdv$, because the induced emf opposes the emf of the battery. The force on the rod is related to the current and the velocity:



ANS. FIG. P31.81

$$F = m\frac{dv}{dt} = IBd$$

$$\frac{dv}{dt} = \frac{IBd}{m} = \frac{Bd}{mR}\left(\mathcal{E} + \mathcal{E}_{\text{induced}}\right) = \frac{Bd}{mR}\left(\mathcal{E} - Bvd\right)$$

To solve the differential equation, let $u = \mathcal{E} - Bvd \quad \rightarrow \quad \frac{du}{dt} = -Bd\frac{dv}{dt}$:

$$\frac{dv}{dt} = \frac{Bd}{mR}\left(\mathcal{E} - Bvd\right)$$

$$-\frac{1}{Bd}\frac{du}{dt} = \frac{Bd}{mR}u \quad \rightarrow \quad \int_{u_0}^{u}\frac{du}{u} = -\int_0^t \frac{(Bd)^2}{mR}dt$$

Integrating from $t = 0$ to $t = t$ gives $\ln\frac{u}{u_0} = -\frac{(Bd)^2}{mR}t$ or $\frac{u}{u_0} = e^{-B^2d^2t/mR}$.

Since $v = 0$ when $t = 0$, $u_0 = \mathcal{E}$; substituting $u = \mathcal{E} - Bvd$ gives

$$\mathcal{E} - Bvd = \mathcal{E}\, e^{-B^2d^2t/mR}$$

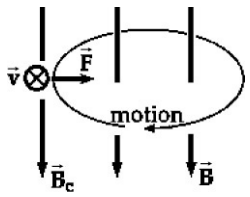
Therefore, $\boxed{v = \frac{\mathcal{E}}{Bd}\left(1 - e^{-B^2d^2t/mR}\right)}$.

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P31.82 Suppose the magnetic field is vertically down. When an electron is moving away from you the force on it is in the direction given by

$$q\vec{\mathbf{v}} \times \vec{\mathbf{B}}_c \quad \text{as} \quad -(\text{away}) \times (\text{down}) = -\,\text{(left)} = (\text{right})$$

Therefore, the electrons circulate clockwise.



ANS. FIG. P31.82

(a) As the downward field increases, an emf is induced to produce some current that in turn produces an upward field to oppose the increasing downward field. This current is directed counterclockwise, carried by negative electrons moving clockwise. Therefore the electric force on the electrons is clockwise and the original electron motion speeds up.

(b) At the circumference, we have

$$\sum F_c = ma_c \quad \rightarrow \quad |q| vB_c \sin 90° = \frac{mv^2}{r} \quad \rightarrow \quad mv = |q| rB_c$$

where B_c is the magnetic field at the circle's circumference.

The increasing magnetic field $\vec{\mathbf{B}}_{av}$ in the area enclosed by the orbit produces a tangential electric field according to

$$\left| \oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} \right| = \left| -\frac{d}{dt} \vec{\mathbf{B}}_{av} \cdot \vec{\mathbf{A}} \right|$$

or

$$E(2\pi r) = \pi r^2 \frac{dB_{av}}{dt} \quad \rightarrow \quad E = \frac{r}{2} \frac{dB_{av}}{dt}$$

Using this expression for E, we find the tangential force on the electron:

$$\sum F_t = ma_t \quad \rightarrow \quad |q| E = m \frac{dv}{dt}$$

$$|q| \frac{r}{2} \frac{dB_{av}}{dt} = m \frac{dv}{dt}$$

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If the electron starts at rest and increases to final speed v as the magnetic field builds from zero to final value B_{av}, then integration of the last equation gives

$$|q|\frac{r}{2}\int_0^{B_{av}}\frac{dB_{av}}{dt}dt = m\int_0^{v}\frac{dv}{dt}dt \quad \rightarrow \quad |q|\frac{r}{2}B_{av} = mv$$

Thus, from the two expressions for mv, we have

$$|q|\frac{r}{2}B_{av} = mv = |q|rB_c \rightarrow B_{av} = 2B_c$$

P31.83 For the suspended mass, M:

$$\sum F = Mg - T = Ma$$

For the sliding bar, m:

$$\sum F = T - I\ell B = ma, \quad \text{where} \quad I = \frac{\mathcal{E}}{R} = \frac{B\ell v}{R}$$

Substituting the expression for current I, the first equation gives us

$$Mg - \frac{B^2\ell^2 v}{R} = (m+M)a \quad \rightarrow \quad a = \frac{dv}{dt} = \frac{Mg}{m+M} - \frac{B^2\ell^2 v}{R(M+m)}$$

The above equation can be written as

$$\int_0^{v}\frac{dv}{(\alpha-\beta v)} = \int_0^{t}dt \quad \text{where} \quad \alpha = \frac{Mg}{M+m} \quad \text{and} \quad \beta = \frac{B^2\ell^2}{R(M+m)}$$

Integrating,

$$\int_0^{v}\frac{dv}{(\alpha-\beta v)} = \int_0^{t}dt \quad \rightarrow \quad \frac{-1}{\beta}\ln(\alpha-\beta v)\Big|_0^{v} = t$$

Then,

$$\left[\ln(\alpha-\beta v) - \ln(\alpha)\right] = -\beta t$$

Solving for v gives

$$\ln\frac{(\alpha-\beta v)}{\alpha} = -\beta t \quad \rightarrow \quad 1 - \frac{\beta}{\alpha}v = e^{-\beta t}$$

$$v = \frac{\alpha}{\beta}\left(1-e^{-\beta t}\right) = \boxed{\frac{MgR}{B^2\ell^2}\left[1-e^{-B^2\ell^2 t/R(M+m)}\right]}$$

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P31.2 (a) Each coil has a pulse of voltage tending to produce counterclockwise current as the projectile approaches, and then a pulse of clockwise voltage as the projectile recedes; (b) 625 m/s

P31.4 +9.82 mV

P31.6 2.26 mV

P31.8 160 A

P31.10 1.89×10^{-11} V

P31.12 (a) $\frac{\mu_0 n\pi r_2^2}{2R}\frac{\Delta I}{\Delta t}$; (b) $\frac{\mu_0^2 n\pi r_2^2}{4r_1 R}\frac{\Delta I}{\Delta t}$; (c) left

P31.14 $\mathcal{E} = -\left(1.42 \times 10^{-2}\right)\cos(120t)$, where t is in seconds and $\mathcal{E}$ is in V

P31.16 $\mathcal{E} = 68.2e^{-1.60t}$, where t is in seconds and $\mathcal{E}$ is in mV

P31.18 (a) See P31.18(a) for full explanation; (b) The emf induced in the coil is proportional to the line integral of the magnetic field around the circular axis of the toroid. Ampère's law says that this line integral depends only on the amount of current the coil encloses.

P31.20 (a) 0.013 3 A; (b) The current is counterclockwise in the lower loop and clockwise in the upper loop.

P31.22 (a) to the right; (b) out of the page; (c) to the right

P31.24 (a) 11.8 mV; (b) The wingtip on the pilot's left is positive; (c) no change; (d) No. If you try to connect the wings to a circuit containing the light bulb, you must run an extra insulated wire along the wing. In a uniform field the total emf generated in the one-turn coil is zero.

P31.26 1.00 m/s

P31.28 $\frac{Rmv}{B^2\ell^2}$

P31.30 The speed of the car is equivalent to about 640 km/h or 400 mi/h, much faster than the car could drive on the curvy road and much faster than any standard automobile could drive in general.

P31.32 $\frac{mgR\sin\theta}{B^2\ell^2\cos^2\theta}$

P31.34 (a) 0.729 m/s; (b) counterclockwise; (c) 0.650 mW; (d) Work is being done by the external force, which is transformed into internal energy in the resistor.

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P31.36 (a) $\frac{N^2B^2w^2v}{R}$ to the left; (b) 0; (c) $\frac{N^2B^2w^2v}{R}$ to the left again

P31.38 (a) 6.00 μT; (b) Yes. The magnitude and direction of the Earth's field varies from one location to the other, so the induced voltage in the wire changes. Furthermore, the voltage will change if the tether cord or its velocity changes their orientation relative to the Earth's field; (c) Either the long dimension of the tether or the velocity vector could be parallel to the magnetic field at some instant.

P31.40 (a) 2.81×10^{-3} N/C; (b) tangent to the electric field line passing through at point P_2 and counterclockwise

P31.42 (a) 12.6 mV; (b) when the plane of the coil is parallel to the magnetic field

P31.44 (a) amplitude doubles and period is unchanged; (b) doubles the amplitude and cuts the period in half; (c) amplitude unchanged and period is cut in half

P31.46 (a) 1.60 V; (b) zero; (c) no change in either answer; (d) See ANS. FIG. P31.46(d); (e) See ANS. FIG. P31.46(e).

P31.48 (a) 110 V; (b) 8.53 W; (c) 1.22 kW

P31.50 See P31.50 for full explanation.

P31.52 $\sim 10^{-4}$ V

P31.54 (a) increasing; (b) 62.6 mT/s

P31.56 $\frac{AB_{\max}}{\tau} e^{-t/\tau}$

P31.58 (a) $\frac{B\ell v}{R}$; (b) particle in equilibrium; (c) 281 m/s; (d) 1.88 A; (e) 169 W; (f) 169 W; (g) yes; (h) increase; (i) yes; (j) larger

P31.60 3.92 m/s is the highest speed the loop can have while the upper edge is above the field, so it cannot possibly be moving at 4.00 m/s.

P31.62 (a) See P31.62(a) for full explanation; (b) 0.250 T

P31.64 (a) 2.54×10^5 m/s; (b) 215 V

P31.66 (a) 0.900 A; (b) 0.108 N; (c) Point *b*; (d) no

P31.68 See P31.68 for full explanation.

P31.70 (a) We would need to know if the field is increasing or decreasing; (b) 248 $\mu\Omega$; (c) Higher resistance would reduce the power delivered.

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P31.72 (a) $C\pi a^2 K$; (b) upper plate; (c) The changing magnetic field through the enclosed area of the loop induces a clockwise electric field within the loop, and this causes electric force to push on charges in the wire

P31.74 See P31.74 for full explanation.

P31.76 $\dfrac{\mu_0 I \ell v}{2\pi R r}\dfrac{w}{(r+w)}$

P31.78 (a) $\dfrac{\left(1.18\times10^{-4}\right)t}{0.800-4.90t^2}$; (b) zero; (c) infinity; (d) 98.3 μV

P31.80 (a) $\dfrac{\pi B_{\max}^2 R^4 b\omega^2}{16\rho}$; (b) 4; (c) 4; (d) 16

P31.82 (a) See P31.82(a) for full description; (b) See P31.82(b) for full description.

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